Available online at www.sciencedirect.com

SCIENCE DIRECT®

Journal of Human Evolution 44 (2003) 401–429

Expansion of the neocerebellum in Hominoidea

Carol E. MacLeod[a,*], Karl Zilles[b,c], Axel Schleicher[b], James K. Rilling[d], Kathleen R. Gibson[e]

[a] *Department of Anthropology, Langara College, 100 West 49 Avenue, Vancouver, BC, V5Y 2Z6 Canada*
[b] *C. & O. Vogt Institut für Hirnforschung, Universität Duesseldorf, D-40001 Duesseldorf, Germany*
[c] *Institute of Medicine, Juelich Research Centre, D-52425 Juelich, Germany*
[d] *Department of Psychiatry and Behavioral Sciences, Emory University, Atlanta, GA 30322, USA*
[e] *Department of Basic Sciences, Dental Branch, DBB 4.133, University of Texas, Houston, TX 77030, USA*

Received 29 April 2002; accepted 25 November 2002

Abstract

Technological and conceptual breakthroughs have led to more serious consideration of the cerebellum as an essential element in cognition. Recent studies show the lateral cerebellum, seat of the neocerebellum, to be most active in cognitive tasks. An examination of the relative volumes of the cerebellar hemispheres in anthropoids would reveal whether some groups show greater neocerebellar development through hemispheric expansion beyond expected allometry, implying a greater contribution of the lateral hemispheres to cognition.

This study expands the existing data on primate brain and brain part volumes by incorporating data from both magnetic resonance scans and histological sections for a total sample size of 97 specimens, including 42 apes, 14 humans and 41 monkeys. The resulting volumes of whole brain, cerebellum, vermis, and hemisphere enable a reliable linear regression contrast between hominoids and monkeys, and demonstrate a striking increase in the lateral cerebellum in hominoids.

The uniformity of the grade shift suggests that this increase took place in the common ancestor to the hominoids. The importance of the neocerebellum in visual-spatial skills, planning of complex movements, procedural learning, attention switching, and sensory discrimination in manipulation would facilitate the adaptation of these early hominoids to frugivory and suspensory feeding.

Keywords: Cerebellum; Vermis; Hemisphere; Neocerebellum; Histological; Magnetic resonance; Anthropoid; Hominoid; Evolution; Cognition

Introduction

The cerebellum is increasingly seen as an important participant in cognitive activities once thought to be the exclusive purview of the

* Corresponding author
E-mail address: caroleli@sfu.ca (C.E. MacLeod).

0047-2484/03/$ - see front matter © 2003 Elsevier Science Ltd. All rights reserved.
doi:10.1016/S0047-2484(03)00028-9

neocortex. More than a device for motor co-ordination and control, the cerebellum is active in sensory discrimination and the integration of sensory and motor functions at a fundamental level (Gao et al., 1996), the planning of complex movements (Thach, 1996, 1997), procedural learning (Doyon, 1997, Fiez and Raichle, 1997, Molinari et al., 1997), visuo-spatial problem solving (Kim et al., 1994), visual attention shifting (Akshoomoff and Courchesne, 1992, Courchesne et al., 1994, Allen et al., 1997), and even language in humans (Raichle et al., 1994). Although motor functions may be evident throughout the cerebellum, these studies show participation in cognitive activities to be concentrated in the lateral cerebellum (hemispheres), whereas basic motor functions such as balance and co-ordination are represented in the medial parts of the cerebellum (vermis and paravermis). Afferent information to the lateral cerebellar cortex can come from both peripheral and central nervous systems but the outgoing information is directed very specifically to the dentate nucleus, from whence much of it travels to the neocortex via the thalamus. The neocerebellum is the localized functional area which dominates the hemispheres in higher primates. Since each zone of the cerebellum is tied to a specific output nucleus, it is feasible to make inferences from volumes of specific cerebellar cortical zones and nuclei with their specific functions. Thus, differential volumetric expansion of the lateral over the medial cerebellum in brain evolution would imply increasing selection for cognitive over purely motor functions.

As part of a larger project in which the volumes of major elements in cerebellar circuitry are taken as an indication of relative functional importance, cerebellar hemisphere volumes were measured in a total sample of 97 monkeys, apes and humans to determine if there was any differential increase in hemisphere size in the higher primates which could, in turn, imply an increase in cerebellar participation in cognition.

The data base for brain and brain-part volumes of Stephan and colleagues (Stephan and Andy, 1969, Stephan et al., 1970, 1981, 1988) forms the foundation for most analyses of brain patterning in the primates, but the data for apes are incomplete. A landmark study by Matano and colleagues measured the deep cerebellar nuclei, the inferior olivary complex, the ventral pons, and the vestibular complex from the specimens in the Stephan collection (Matano et al., 1985a,b; 1986, 1992). The present study expands the data available on apes in particular, adding the orangutan and the bonobo, and increasing the number of ape and human specimens to permit reliable analysis with linear regression in hominoid/monkey or ape/human/monkey contrasts. Rather than smooth the data to species means, individual data points have been used to allow the regression model to incorporate biological variability within the sample of each taxon.

The presentation of the volumes in a statistical context enables a deeper understanding of intraspecific and species variability within larger taxonomic categories. This will serve as a reference to other scientists who discern evolutionary trends in brain expansion with unavoidably small sample sizes of extant primate brains and fossil endocasts.

Material and methods

The volumes reported in this study are drawn from two primate brain collections. One consists of 47 *in vivo* magnetic resonance brain scans compiled by the Yerkes Regional Primate Research Center in Atlanta, Georgia ("Yerkes sample"). This sample of 11 species of primates includes *Saimiri sciureus, Cebus apella, Papio cynocephalus, Cercocebus torquatus atys, Macaca mulatta, Hylobates lar, Pongo pygmaeus, Gorilla gorilla, Pan paniscus, Pan troglodytes*, and *Homo sapiens sapiens*. The collection from the Institut für Hirnforschung in Duesseldorf, Germany ("Hirnforschung sample") contains histological, serially sectioned primate brains. This sample of 50 brains encompasses 16 species of anthropoids, including the hominoid species of *Hylobates lar, Pongo pygmaeus, Gorilla gorilla, Pan paniscus*, and *Pan troglodytes*, as well as *Erythrocebus patas, Macaca mulatta, Cercopithecus* (species unknown), *Cercocebus albigena, Papio cynocephalus, Ateles paniscus, Alouatta seniculus, Cebus* (species unknown), *Saimiri sciureus*, and *Aotus trivirgatus*. These specimens are drawn from the collection of one of the authors (K.Z.), although three chimpanzees, one gorilla and one gibbon are from the "Stephan collection" housed at the Institut für Hirnforschung. Two of the Duesseldorf specimens were not suitable for measurement of the vermis. The combined Yerkes and Hirnforschung samples contain 42 ape brains, the largest number of ape brain and brain part volumes ever assembled.

Yerkes sample

In the scanning procedure carried out by one of the authors (J.K.R), all subjects were treated according to the protocols established by Emory University. Prior to scanning, nonhuman primates were anaesthetized with Ketamine (10 mg/kg) and then weighed. Monkeys and gibbons were scanned in a prone position with their heads in a human knee coil, while the larger hominoids were scanned in a supine position using the human head coil. Throughout the scan, nonhuman primates were anaesthetized with a continuous intravenous infusion of propofol (10–20 mg/kg/hr). T1-weighted images of the entire brain were acquired with a 1.5 Tesla Philips NT scanner (Philips Medical systems, The Netherlands) using a gradient-echo protocol with the following parameters: slice thickness = 1.2 mm, slice interval = 0.6 mm, TR = 19.0 msec, TE = 8.5 msec, number of signals averaged = 8, matrix = 256 × 256 pixels. Field of view (FOV) was adjusted to include the entire brain in each subject, and varied as a function of brain size, ranging from 120 mm in squirrel monkeys to 200 mm in humans. Consequently, pixel size ranged from 0.47 mm^2 to 0.78 mm^2. Despite pixel size being smaller in small brains, larger brains had more total pixels than smaller brains, and therefore better resolution relative to brain size. Approximate scanning time for monkeys, apes and humans was 40 minutes, 60 minutes and 80 minutes respectively.

All measurements were made with "Easy Vision", an image-analysis software programme (Philips Medical Systems, The Netherlands). Whole brain volumes were compiled in the horizontal plane by J.K.R. (Rilling and Insel, 1998), and the cerebellum and vermis were measured in sagittal view by C.E.M. (MacLeod, 2000). For the cerebellar and vermis measurements, the scans were first reformatted in sagittal view along the anterior-posterior commissure axis (Talairach and Tournoux, 1988), then sliced as thinly as possible (0.5 mm.) to facilitate accurate contour tracing. All slices in the image stack were measured (from 12 to 30 slices for the vermis, and from 36 to 210 slices for the cerebellum). Brain tissue was segmented from meninges and bone by setting the intensity threshold to "saturate" brain tissue only according to the grey-value intensity of each pixel. The observer outlined the vermis with a flexible "lasso" to distinguish it from other brain tissue, and kept cerebellar and vermis intensity thresholds the same for each specimen to minimize measurement error (MacLeod, 2000).

One variable that is often ignored in MR morphometry is intensity threshold. Since it is set interactively by visually determining the correspondence between the segmentation result and the outline of tissue, the subjective judgement of setting the intensity threshold may be highly variable, and the volumes may differ substantially. The first author measured 7 MR-scanned cerebellum and vermis volumes with the lowest and highest thresholds that would adequately detect the tissue. There was a surprisingly wide range of possible volumes between the high and low thresholds, with the coefficients of variation (CV) in the cerebellum ranging from 3.7% to 20.9%, and in the vermis ranging from 5.6% to 27.7%. The average CV between high and low thresholds was 10.3% in the cerebellum, and 9.7% in the vermis. Thus, to reduce variation in measurement, the threshold intensity for both the measurement of the vermis and the cerebellum was kept constant for each scan. A sample-wide threshold is impossible because of the variation inherent in each MR scan. It is probable that this precaution of controlling for intensity threshold within scans accounted largely for a 41% reduction in intra-observer error from measures taken in 1999 over the first measures taken in 1997.

Cerebellar hemisphere volumes were obtained by subtracting the volume of the vermis from that of the cerebellum. This study identified and

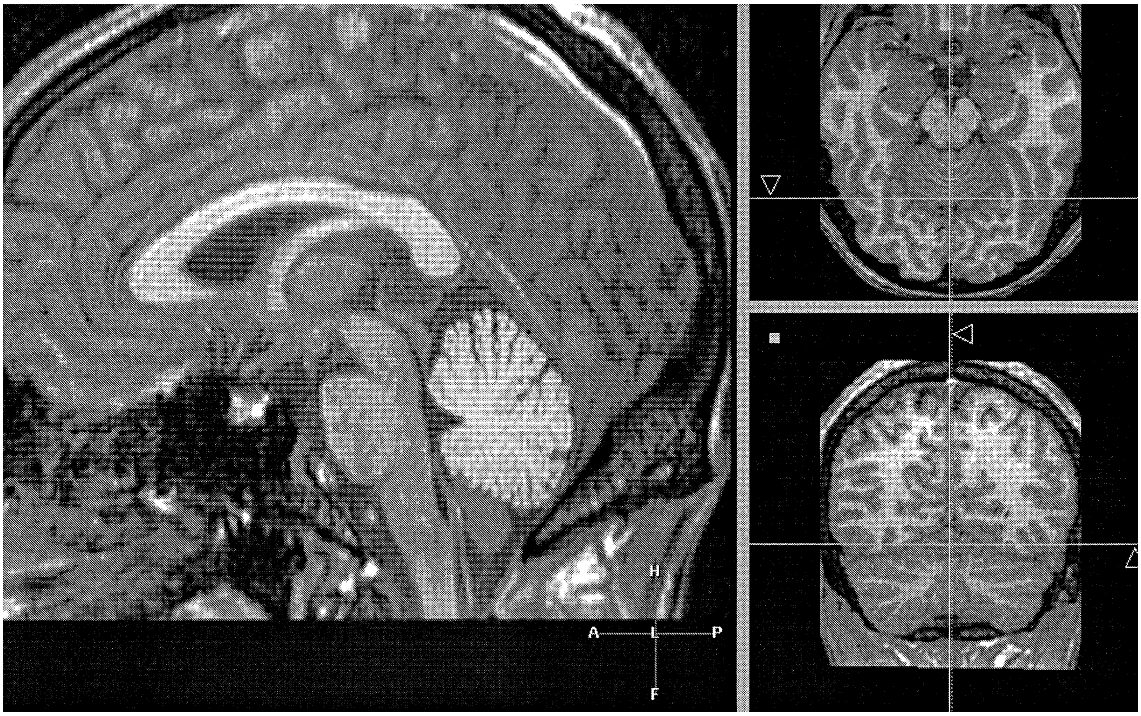

Fig. 1. Magnetic resonance image analysis screen showing sagittal, horizontal and coronal views of the vermis.

measured the vermis by using the three-dimensional nature of MR scans. While the main screen showed the area of interest in the sagittal plane, two other screens showed the vermis in the coronal (frontal) and horizontal planes. When the mouse was clicked on a particular spot in one view, its exact location was shown in the other two views. This way, the accuracy of the selected area could be verified along the three dimensions.

The folia of the vermis emerge orthogonal to the sagittal plane, whereas the folia of the hemispheres are aligned along the sagittal plane in a medial to lateral direction. The boundaries of the vermis were set at this change of orientation, marked in the gross anatomy of the cerebellum by the paramedian sulcus dorsally. The exact point of reorientation of the folia from vermis to hemisphere was sometimes difficult to discern, but practiced consistency in the decision-making process led to a reasonably low intra-observer error (Table 3). No gross anatomical distinction can be made between the paravermis and neocerebellum, but microscopic studies of ontogenetic development indicate that the neocerebellum dominates the cerebellar hemispheres in higher primates (Jansen and Brodal, 1954). Thus, an increase in the lateral cerebellum is interpreted as an increase in the neocerebellum.

Hirnforschung sample

Most brains of the Hirnforschung sample were serially sectioned in the coronal plane and stained for cell bodies with a silver-staining method after Gallyas (Merker, 1983) (see Table 2 for further details). In some of the older brains in the Hirnforschung and Stephan collections, Nissl cresyl-violet staining for cell bodies was used in place of the silver Gallyas staining. If staining intensity in any specimens had decayed with age, then myelin-stained sections were used by default.

An average of 25 equally spaced sections per whole brain and cerebellum were digitized by a

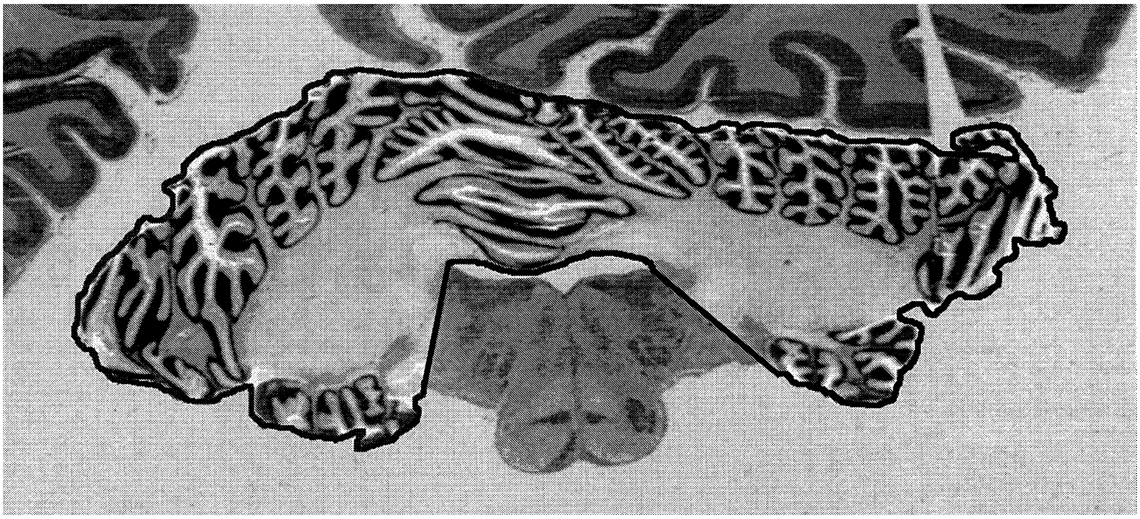

Fig. 2. Orangutan cerebellum defined interactively using the KS400© system. The computer records the area inside the boundary line. In some views of the cerebellum, cerebellar white matter runs continuously with the peduncles. Rather than include all of the peduncles with cerebellar volume, a "boundary" between the two is determined in this case by a straight line drawn between the folia and the lateral reaches of the fourth ventricle. Figs. 2, 3 and 4 have been enhanced for black and white photographs with Adobe Photoshop.

CCD camera and stored in the computer. The number of sections per specimen guaranteed a high precision in the volumetric calculation (Zilles et al., 1982). The author (C.E.M.) corrected for the camera magnification interactively, outlined the area of interest in each section with the mouse, and adjusted the threshold intensity to conform to tissue boundaries. Since the intensity of the histological stains varied among sections, it was not feasible to adopt one single threshold intensity for the whole set of sections. The areas for each structure were then summed and multiplied by the distance between each section (after Cavalieri, in Howard and Reed, 1998). The whole brains of the collection were fixed in formalin or Bouin's fluid, dehydrated in a series of alcohol solutions of increasing concentration (70%–99%), embedded in paraffin, then serially sectioned to a thickness of 20 μm using a microtom. Dehydration and embedding in hot paraffin result in considerable shrinkage (Stephan et al., 1981), but the native volumes before histological processing can be calculated with a correction factor ("shrinkage factor"), established by comparing the *post mortem* brain volume to the fixed brain volume. The *post mortem* brain volume was calculated using the fresh weight of the brain divided by the specific gravity of brain tissue (1.036 grams/cm^3, in Stephan et al., 1981). To calculate absolute volumes for the purposes of comparison, the correction factor was applied to all structures measured from that particular brain. In cases where *post mortem* weight was not known, the average correction factor for the sample was applied.

Stephan and colleagues (1981) included both the peduncles and basal pons in the measure of cerebellar volume, but the protocols used in our study excluded the peduncles at the borders of the cerebellar cortex, and did not include any brain stem structures in the calculation of cerebellar volume, as illustrated in Fig. 2[1]. Pontine volumes are very small in comparison to total cerebellar volume (see Matano et al., 1985a); thus cerebellar volumes, especially in the larger brained primates, are comparable between the Stephan sample and our study.

As with the MR scans, the hemisphere volume was obtained by subtracting the vermis from the

[1] A disk with examples of cerebellum and vermis measurement protocols can be sent to specialists upon request to the first author.

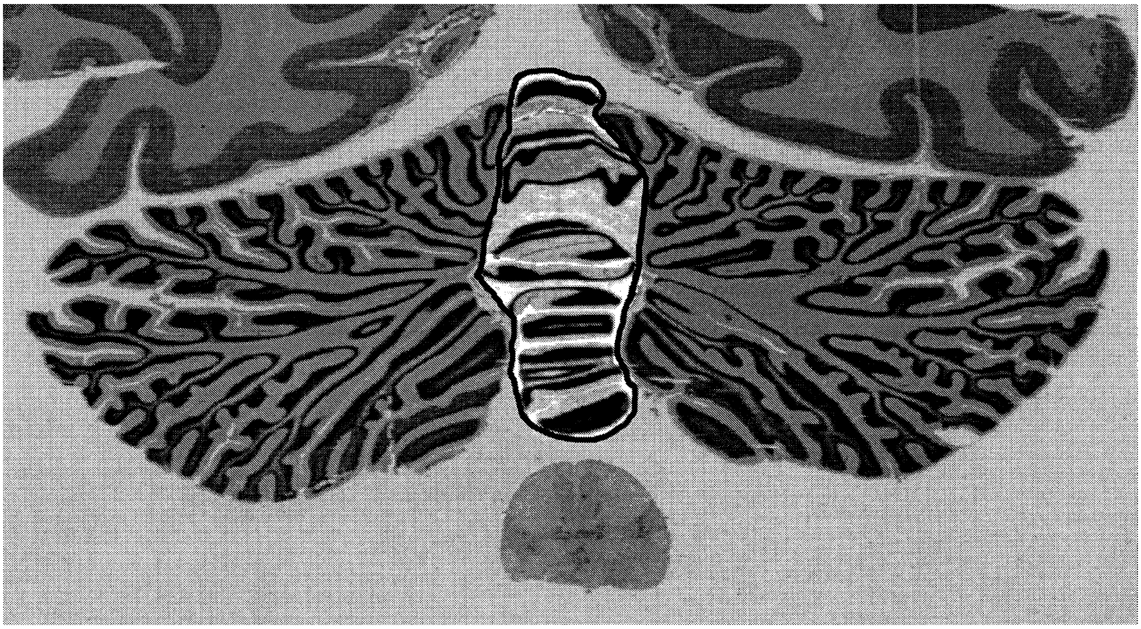

Fig. 3. Gibbon cerebellum showing the vermis outlined in black.

cerebellum volume. The measurement of the vermis in the histological sections could not take advantage of the three-dimensional method used in the MR scans, but did use protocols that corresponded as closely as possible to boundaries established in the coronal plane in the MR scans. The boundary between hemisphere and vermis was established according to folial orientation, as described above for the MR scans, but the histological sections afforded much higher resolution than the scans (Figs. 3 and 4). The flocculus, a small part of the vermis, extends beyond the midline. It is impossible to distinguish from the hemispheres in the MR sections, and difficult in the histological sections, but is too small in the anthropoids to affect vermis volume estimates significantly.

To establish species-specific standards for brain and brain-part volumes, Stephan et al., (1981) corrected their measured volumes to standard brain weights established by Bauchot and Stephan, (1966, 1969) by multiplying each volume by an "average correction factor" or C_{sta} (Stephan et al., 1981). The present study did not standardize the data but instead increased the sample size within hominoid taxa to provide reliable means.

Statistical analysis

All statistical analyses were done with the Systat 8 programme, and the charts were generated using Microsoft Excel (v.X).

The data were treated from many statistical perspectives, but the most telling results came from the application of multiple linear regression with a discrete variable, a form of ANCOVA (Kleinbaum and Kupper, 1978). With this method, the data could be subdivided to test whether "x" could better predict "y" using subgroups rather than the sample as one grouping. The subgroups in this study were monkeys and hominoids, or monkeys and apes, with humans placed independent of the regression line. These groupings were termed "grades" and imply different levels of complexity in the exploitation of resources (Huxley, 1958). The statistical method enabled the validity of these groupings to be tested to determine whether a single regression line or two regression lines based on grade were the best fit. A grade shift would be expressed as a break in the expected scaling principle, shown in the graphs of the regressions by the vertical

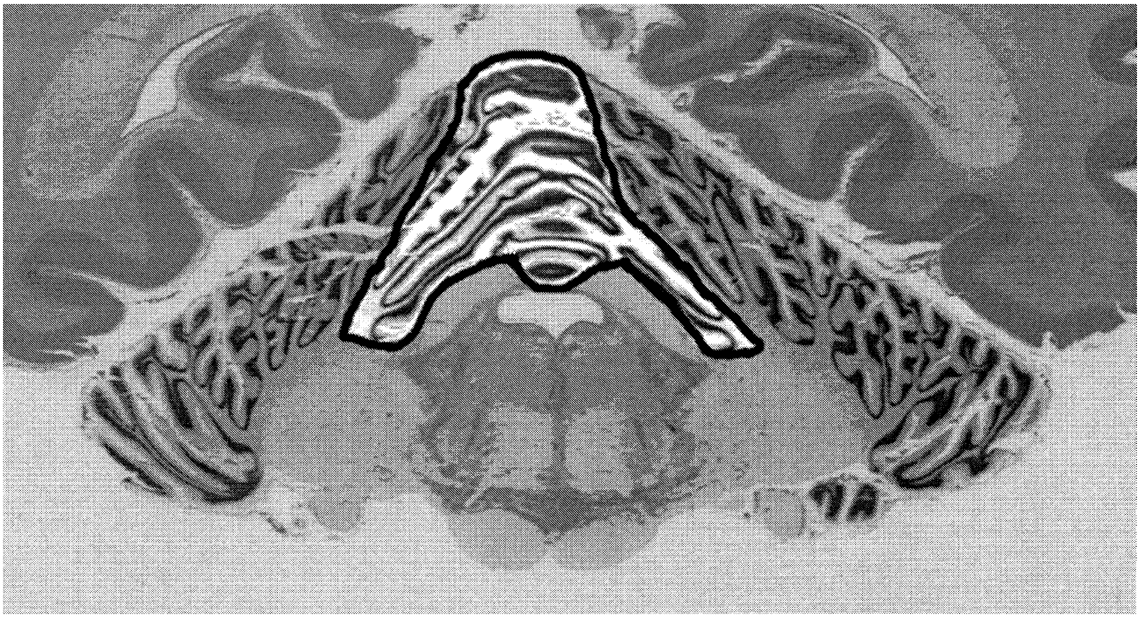

Fig. 4. Vermis in an *Ateles* cerebellum. Folia are counted as vermal if they cross the midline. Superior cerebellar peduncles are cut out; otherwise medial white matter with fastigial and interposed nuclei is included.

separation of parallel lines (Martin, 1980, Martin and Harvey, 1985).

Grade, a qualitative variable, must be entered as a "dummy variable" in the equation, in which all hominoids are given the value of 1, and all monkeys the value of 0. The basic equation is as follows (Kleinbaum and Kupper, 1978):

$$y' = \beta_0 + \beta_1 x_1 + \beta_2 x_2$$

Where:
y' = predicted volume
β_0 = constant
x_1 = predictor volume
x_2 = grade (1 if hominoid, 0 if monkey)
β_1 and β_2 = coefficients of x_1 and x_2 respectively

This basic model graphs the two regression lines as parallel lines with the same slopes that are the average of the slopes for monkeys and for hominoids. If x_1 and x_2 are allowed to interact, however, then the true slopes of β_1 and β_2 for each grade will be shown, with p-values of the interaction variable indicating whether the slopes differ significantly (i.e., when p is well above 0.05, the slopes are considered to be parallel). With this method, the observer does not impose an artificial construct on the data, but tests the validity of both grade and slope, targeting the actual patterning of allometric increase. Sometimes this model is too sensitive because of co-linearity and the high correlation between the two x-values and y, in which case the question of a grade shift should be resolved by using both the basic model and the model with interaction. The equation for the full model with interaction is: $y' = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1 x_2$ (Kleinbaum and Kupper, 1978). This differs from the basic formula in the interaction of both volume and grade, with β_3 as their coefficient.

Although treatment of the data by major axis and reduced major axes can include the error of the x-axis in the equation, and thereby give a more accurate slope to the data (Harvey and Mace, 1982), differences among the methods are minor when correlation coefficients are high (Aiello, 1992), as is the case in the regressions in this study. With only six genera in *Hominoidea*, it is difficult to obtain an accurate regression line for this grade if means alone are used. Both the gibbons and humans exercise leverage, and one aberrant mean can significantly affect the slope. If the individual data points are used instead, increasing the degrees of freedom, one or two extreme values out of 42 will not have the same impact on the regression line as 1 of 6 points. Examination of slopes of all volumes in the entire project show slope to be affected by the breadth of species represented in a given sample[2], and until this effect is better understood, slope *per se* is not given undo interpretive weight.

Independent contrasts

Linear regression requires that all data points be independent. This is not the case when regression is applied to a phylogenetic series, with its hierarchically nested clades. The method of comparative analysis by independent contrasts, or CAIC (Purvis and Rambaut, 1995), was applied to the data set to test whether the patterns revealed by multiple regression would remain in an analysis that incoporated phylogenetic contrasts. In this method, contrasts between closely related species, such as *Pan troglodytes* and *Pan paniscus*, are established for x and y variables, and this value is then contrasted with Pan's closest relative, *Homo*, with this group contrasted against the next closest relative, the gorilla, etc. Data points are thus compared in a phylogenetically consistent way, with branch lengths incorporated into the analysis (from the phylogeny of Purvis, 1995). The independent contrasts method can establish a more reliable picture than conventional regression when spurious correlations resulting from the nonindependence of species are eliminated (Purvis and Rambaut, 1995, Nunn and Barton, 2001).

Results

The volumes of the whole brain, cerebellum, vermis, and cerebellar hemispheres in the Yerkes and Hirnforschung samples are reported in Tables 1 and 2. Intra- and inter-observer error are shown in abbreviated form in Table 3. Table 4 presents the ranges, medians, means, standard deviations and coefficients of variation in the two samples in order to convey variability of primate brain and brain-part volumes, and Table 5 compares the Yerkes and Hirnforschung data.

Variability and error in measured volumes

Before examining the substance of the results, it is important to situate the volumes in a statistical context.

Table 3 shows that intra-and inter-observer error, expressed as a coefficient of variation (CV), were reasonable in both samples. However, the error in the Yerkes sample was higher. This may be due to the low resolution of the scans in comparison to the histological sections, and to the more precise measuring procedure of the KS400® software programme developed at the Brain Research Institute, Duesseldorf.

Table 4 shows the large range of variability within a given primate species for both the Yerkes and Hirnforschung data sets. In the case of gorillas, orangutans, humans, and to a smaller extent, chimpanzees, much of this variability can be explained by brain dimorphism between males and females. The male gorillas in the combined sample for whom sex is known (n = 2), for example, have a mean brain volume of 449.2 cm^3, whereas the females have a mean brain volume of 346.8 cm^3 (n = 2). Age at death is another factor. The Yerkes sample uses only adult animals, but the Hirnforschung sample has primates with a wider age-range. This could account for some higher standard deviations in the Hirnforschung sample than in the Yerkes, yet the average CV's for grade are higher for Yerkes than for Hirnforschung. Wide variation in species volumes is evident in both data sets, and is simply a consequence of the biological variability of brain and brain-part volumes within a primate species.

Table 5 compares the Yerkes and Hirnforschung data sets for taxa in common. Most of the comparisons are at the level of species, although *Cercocebus* and *Cebus* are at the level

[2] Generally, the slope is not as steep if small-brained monkeys are not included. In some cases, monkeys with brains under 30 cm^3 form a parallel grouping with the rest of the anthropoids, including humans (MacLeod, 2000), and this phenomenon should be investigated further.

Table 1
Volumetric data from the Yerkes sample

Specimen	Sex	Brain volume cm^3	Cerebellum volume cm^3	Vermis volume cm^3	Hemisphere volume cm^3
Homo sapiens 1	F	1088	111.4	9.3	102.1
Homo sapiens 2	M	1212	119.6	9.4	110.2
Homo sapiens 3	F	1330	148.2	11.6	136.6
Homo sapiens 4	F	1334	147.2	14.1	133.1
Homo sapiens 5	M	1403	141.7	11.6	130.1
Homo sapiens 6	M	1426	154.0	15.0	139.0
Gorilla gorilla (Kinyani)	F	330.7	56.1	4.9	51.2
Gorilla gorilla (Kekla)	M	463.9	81.2	9.1	72.1
Pan paniscus (Brian)	M	322.4	46.1	6.1	40.0
Pan paniscus (Bosandjo)	M	331.8	51.4	9.6	41.8
Pan paniscus (Lorel)	F	309.2	39.7	4.9	34.8
Pan paniscus (Jill)	F	281.4	39.5	4.8	34.7
Pan troglodytes (Kengee)	F	343.8	50.1	5.3	44.8
Pan troglodytes (Merv)	M	391.2	51.4	5.0	46.4
Pan troglodytes (Jimmy Carter)	M	311.4	51.8	4.9	46.9
Pan troglodytes (Laz)	M	365.9	45.7	4.9	40.8
Pan troglodytes (Lulu)	F	282.7	45.0	6.1	38.9
Pan troglodytes (Mary)	F	328.6	39.6	4.0	35.6
Pan troglodytes (Panzee)	F	317.9	46.1	4.7	41.4
Pongo pygmaeus(Hati)	F	343.0	42.7	4.7	38.0
Pongo pygmaeus(Mentubar)	M	383.2	45.2	5.1	40.1
Pongo pygmaeus(Minyak)	M	423.8	59.3	6.1	53.2
Pongo pygmaeus(Molek)	M	477.5	52.0	4.9	47.1
Hylobates lar (Sunny)	M	76.1	10.4	1.7	8.7
Hylobates lar (Cherokee)	F	99.7	13.5	2.4	11.1
Hylobates lar (Cleo)	F	80.0	13.6	1.9	11.7
Hylobates lar (Buddy)	M	76.3	9.0	1.8	7.2
Papio cynocephalus (B263)	M	159.2	14.8	3.9	10.9
Papio cynocephalus (140M)	M	127.4	12.4	3.3	9.1
Cebus apella (Binkey)	F	60.1	5.4	1.7	3.7
Cebus apella (Diva)	F	55.2	5.1	1.4	3.7
Cebus apella (Andy)	M	73.2	7.7	2.0	5.7
Cebus apella (Vincent)	M	77.3	7.6	2.2	5.5
Macaca mulatta (153C)	M	72.8	6.3	1.8	4.5
Macaca mulatta (RAI5)	M	82.3	7.7	2.8	4.9
Macaca mulatta (REG3)	M	85.4	7.5	2.3	5.2
Macaca mulatta (RUL1)	M	84.4	7.6	2.2	5.4
Macaca mulatta (RPE1)	M	99.5	8.3	3.0	5.2
Macaca mulatta (RAK5)	NA	NA	6.0	1.9	4.1
Cercocebus torquatus atys (FYF)	M	97.9	10.6	2.9	7.7
Cercocebus torquatus atys (FSO)	M	96.2	8.1	2.3	5.9
Cercocebus torquatus atys (FFK)	F	97.8	7.4	1.9	5.5
Cercocebus torquatus atys (FWJ)	M	103.5	8.5	2.1	6.4
Saimiri sciureus (93M)	M	23.6	2.1	1.0	1.1
Saimiri sciureus (93I)	F	23.7	1.9	0.8	1.1
Saimiri sciureus (S104)	M	20.8	2.0	0.8	1.2
Saimiri sciureus (S105)	M	24.3	2.2	0.9	1.2

Table 2
Volumetric data from the Hirnforschung sample

Specimen	Sex	Brain Volume cm^3	Cerebellum Volume cm^3	Vermis Volume cm^3	Hemisphere Volume cm^3
Homo sapiens (139/95)	M	1174.7	125.3	10.6	114.6
Homo sapiens (H9/88)	M	1582.6	182.1	15.7	166.3
Homo sapiens (SN382/81)	F	1102.3	138.2	9.6	128.6
Homo sapiens (71/84) *	F	1093.6	133.2	12.3	120.9
Homo sapiens (14/94)	F	1137.2	136.9	14.7	122.1
Homo sapiens (146/86)	M	1387.1	138.6	9.5	129.0
Homo sapiens (SN201/86) *	M	1176.6	132.0	9.9	122.1
Homo sapiens (503/81) *	F	1264.5	139.8	11.2	128.7
Gorilla gorilla (A375) †	M	434.4	60.1	7.2	52.9
Gorilla gorilla (1001)	NA	414.5	51.0	4.5	46.5
Gorilla gorilla (YN82-140)	F	362.9	48.5	5.8	42.8
Pan paniscus (YN86-137)	F	378.4	54.1	5.7	48.3
Pan paniscus (Zahlia 1/97)	F	312.7	53.0	7.8	45.2
Pan troglodytes (YN89-278) *	M	405.4	53.5	4.2	49.3
Pan troglodytes (1548) ‡	NA	265.0	28.2	3.7	24.5
Pan troglodytes (PAN 1) †	F	424.7	56.5	7.5	49.0
Pan troglodytes (Bathsheba 4/97)	F	347.0	47.8	6.8	41.0
Pan troglodytes (Schimpanse 280) †	M	445.0	45.3	5.7	39.6
Pan troglodytes (Schimpanse 278) †§	F	405.4	53.3	NA	NA
Pan troglodytes (Schimpanse A269) †*	M	333.5	40.6	3.4	37.2
Pongo pygmaeus (0475)	F	221.0	22.4	3.5	18.8
Pongo pygmaeus (YN85-38)	M	356.2	41.2	NA	NA
Pongo pygmaeus (Harry 2/97)	M	424.7	54.3	5.4	48.9
Pongo pygmaeus (Briggs 5/97)	M	333.0	39.2	3.7	35.5
Hylobates lar (YN81-146)	F	88.8	10.6	2.0	8.6
Hylobates lar (2436) (LH) ‡	NA	107.9	14.5	2.3	12.2
Hylobates lar (1547)	M	79.3	12.3	2.4	9.9
Hylobates lar (Disco 3/97)	F	115.8	13.8	2.3	11.5
Hylobates lar (1203) †	M	98.4	12.0	2.6	9.4
Erythrocebus patas (1545) ‡	M	89.0	7.9	2.7	5.2
Macaca mulatta (Rhesus 516)	F	101.2	9.8	3.3	6.4
Macaca mulatta (Rhesus R4) *‡	NA	108.3	9.4	3.0	6.3
Cercopithecus sp. (S) ‡	NA	57.0	5.6	2.2	3.3
Cercopithecus sp. (CER) ‡	M	71.0	6.8	3.1	3.6
Cercocebus albigena (A221)	M	112.0	11.7	4.0	7.7
Papio cynocephalus (P3)	F	141.9	12.8	3.5	9.3
Papio cynocephalus (P4)	F	150.7	15.3	4.4	11.0
Ateles paniscus (2412) ‡	NA	107.3	10.7	3.9	6.8
Ateles paniscus (1541) ‡	M	93.0	11.7	2.9	8.7
Ateles paniscus (1538) ‡	M	71.8	8.7	2.3	6.4
Alouatta seniculus (1961/8) ‡	M	36.5	4.3	1.6	2.7
Alouatta seniculus (1961/9)	F	38.3	4.0	1.3	2.7
Cebus sp. 1 LH§ RH coronal ‡	M	79.1	7.3	2.5	4.8
Cebus sp. 5 ‡	NA	73.9	6.8	2.2	4.6
Saimiri sciureus (1064)	F	19.0	1.9	0.9	1.0
Saimiri sciureus (2408) ‡	M	20.7	1.9	1.0	0.9
Aotus trivirgatus (1536) ‡	F	18.9	1.7	1.0	1.2
Aotus trivirgatus (K197)	F	15.8	1.7	0.8	0.9
Aotus trivirgatus (888) ‡	NA	19.7	1.8	0.9	0.8
Aotus trivirgatus (1535) ‡	F	22.1	1.8	0.9	0.9

All specimens are sectioned coronally unless otherwise indicated.
[*]Horizontal sections.
[†]Specimens from the Stephan Collection.
[‡]Specimens for which *post-mortem* brain weights are not known.
[§]Sagittal sections.

Table 3
Intra- and inter-observer error

	Intra-observer error				Inter-observer error*			
	Hirnforschung		Yerkes		Semendeferi		Rilling	
Volumes	AV. Error CV	N=	AV. Error CV	N=	AV. Error CV	N=	AV. Error CV	N=
Whole brain	2.83	8	–	–	1.77	4	–	
Cerebellum	1.17	2	2.53	4**			3.71	43
Vermis	2.43	16	4.82	6***				

Intra-and inter-observer error expressed as coefficient of variation.
*Inter-observer error-Measurements of whole brain volume from Hirnforschung compared with same specimens measured by K. Semendeferi using the Cavalieri method. Measurements of cerebellar volume from Yerkes on sagittal plane (C.E.M.) compared with those obtained by J.Rilling on horizontal plane from same scans.
**Cerebellum was measured in 2 different axes, the sagittal and horizontal, for the purposes of comparison.
***After establishment of new protocols in 1999 visit, and standardization of threshold intensity.

of genus. The variation between the Yerkes and Hirnforschung means is much less than the variation within each taxa. When the common species means for whole brain, cerebellum, vermis and hemisphere are compared in a two-tailed, paired t-test, the two data sets are not significantly different. In fact, the Stephan, Yerkes and Hirnforschung sets accord quite well in the mean volumes of whole brain and cerebellum, with no significant differences in the p-values for species and genera in common, and thus present a cross-validation of their reliability. Figs. 5 and 6 show the Yerkes and Hirnforschung means regressed against the Stephan means for the same taxa of whole brain and cerebellum volumes. Even though Stephan and colleagues (1981) used different protocols for measuring the cerebellum, the three data sets are in strong accord, as shown by the two overlapping regression lines with slopes close to 1, intercepts close to 0, and high r^2 values. There is no evidence that the application of a correction factor for shrinkage in the fixed brains results in less reliable volumes than in the *in vivo* MR scans, where no correction factor is necessary.

The homogeneity of the Yerkes and Hirnforschung means justifies their combination into one set, hereafter used for all discussion of results. A larger data set yields more precise estimates of means and standard deviations, as shown by the combined SEM values, and provides a more reliable basis for treatment of the data with multiple regression because of the greater number of degrees of freedom.

Although the means of the three data sets accord with one another, there is still evidence of a high degree of biological variability within each taxa, which would preclude an easy interpretation of species volumes in a socio-ecological context. It is hoped that researchers will be able to use the descriptive statistical data presented here to temper their conclusions on the significance of species differences. Our study concentrates on the larger pattern shown by monkey and hominoid grades; a more finer grained interpretation must await further analysis, after volumes in the complete set are reported.

Ratios and proportions

Table 6 summarizes three important sets of ratios, with standard deviation (SD), and standard error of the mean (SEM) calculated from the mean of the ratios in each grade. A one-way ANOVA applied to the three sets of ratios reported here shows the difference in grade means to be significant at the 0.0001 level. The percentage of cerebellum to whole brain from the combined sample is 11% in humans, 13.4% in great apes, 13.4% in gibbons, 9.2% in Old World monkeys, and 9.8% in New World monkeys. Tukey's HSD multiple comparisons show all ape ratios to be significantly different from monkey ratios, with no significant differences between lesser and greater

Table 4
A. Hirnforschung ranges, medians, means, standard deviations (SD), and coefficients of variation (CV)

		Brain volume						Cerebellum volume				
Number	Genus	Range cm^3	Median cm^3	Mean cm^3	SD cm^3	CV %	Number	Range cm^3	Median cm^3	Mean cm^3	SD cm^3	CV %
8	*Homo*	1102–1583	1175.7	**1239.8**	168.5	13.6	8	125–182	137.5	**140.8**	17.4	12.3
3	*Gorilla*	363–434	414.5	**403.9**	36.9	9.1	3	49–60	51.0	**53.2**	5.0	9.4
2	*Pan paniscus*	313–378	345.6	**345.6**	46.4	13.4	2	53–54	53.5	**53.5**	0.8	1.4
7	*Pan troglodytes*	265–445	405.4	**375.1**	63.0	16.8	7	28–57	47.8	**46.5**	9.7	21.0
4	*Pongo*	221–425	344.6	**333.7**	84.6	25.4	4	23–54	40.2	**39.3**	13.1	33.3
5	*Hylobates*	79–116	98.4	**98.0**	14.6	14.9	5	10.6–14.5	12.3	**12.6**	1.6	12.3
1	*Erythrocebus*	89	89.0	**89.0**	–	–	1	7.9	7.9	**7.9**	–	–
2	*Macaca*	101–108	104.8	**104.8**	5.0	4.7	2	9.4–9.8	9.6	**9.6**	0.3	3.1
2	*Cercopithecus*	57–71	64.0	**64.0**	9.9	15.5	2	5.6–6.8	6.2	**6.2**	0.9	14.1
1	*Cercocebus*	112	112.0	**112.0**	–	–	1	10.4	11.7	**11.7**	–	–
2	*Papio*	142–151	146.3	**146.3**	6.2	4.2	2	12.8–15.3	14.1	**14.1**	1.8	12.8
3	*Ateles*	72–107	93.0	**90.7**	17.9	19.7	3	8.7–11.7	10.7	**10.4**	1.5	14.6
2	*Alouatta*	37–38	37.4	**37.4**	1.3	3.4	2	4.0–4.3	4.1	**4.1**	0.2	5.8
2	*Cebus*	74–79	76.5	**76.5**	3.7	4.9	2	6.8–7.3	7.1	**7.1**	0.3	4.7
2	*Saimiri*	19–21	19.8	**19.8**	1.2	6.2	2	1.9	1.9	**1.9**	0.0	2.1
4	*Aotus*	16–22	20.1	**19.5**	2.7	13.7	4	1.7–1.8	1.8	**1.8**	0.2	11.0
	Ranges, medians, means, standard deviations, and coefficients for grades based on Genus/Species means above											
8	Human	1102–1583	1175.7	**1239.8**	168.5	13.6	8	125–182	137.5	**140.8**	17.4	12.3
16	Great ape	221–445	370.7	**364.6**	31.5	8.6	16	23–60	49.8	**48.1**	6.7	14.0
5	Lesser ape	79–116	98.4	**98.0**	14.6	14.9	5	10.6–14	12.3	**12.6**	1.6	12.3
8	OW monkey	57–151	104.8	**103.2**	30.3	29.4	8	5.4–15	9.6	**9.9**	3.1	31.4
13	NW monkey	16–107	37.4	**49.0**	32.7	66.8	13	1.7–11.7	4.1	**5.1**	3.7	72.1

		Vermis volume						Hemisphere volume				
Number	Genus	Range cm^3	Median cm^3	Mean cm^3	SD cm^3	CV %	Number	Range cm^3	Median cm^3	Mean cm^3	SD cm^3	CV %
8	*Homo*	9.5–15.7	10.9	**11.7**	2.4	20.3	8	114.6–166.3	125.4	**129.0**	15.9	12.3
3	*Gorilla*	4.5–7.2	5.8	**5.8**	1.4	23.3	3	42.8–52.9	46.5	**47.4**	5.1	10.8
2	*Pan paniscus*	5.7–7.8	6.8	**6.8**	1.5	21.6	2	45.2–48.3	46.7	**46.7**	2.2	4.8
6	*Pan troglodytes*	3.4–7.5	4.9	**5.2**	1.7	32.8	6	24.5–49.3	40.3	**40.1**	9.1	22.7
3	*Pongo*	3.5–5.4	3.7	**4.2**	1.0	24.8	3	18.8–35.5	35.5	**34.4**	15.0	43.7
5	*Hylobates*	2.0–2.6	2.3	**2.3**	0.3	14.0	5	8.6–12.2	9.9	**10.3**	1.5	14.5
1	*Erythrocebus*	2.7	2.7	**2.7**	–	–	1	5.2	5.2	**5.2**	–	–
2	*Macaca*	3–3.3	3.2	**3.2**	0.2	6.6	2	6.3–6.4	6.4	**6.4**	0.1	1.3
2	*Cercopithecus*	2.2–3.1	2.7	**2.7**	0.6	0.2	2	3.3–3.6	3.3	**3.5**	0.2	6.8
1	*Cercocebus*	4	4.0	**4.0**	–	–	1	7.7	7.7	**7.7**	–	–
2	*Papio*	3.5–4.4	3.9	**3.9**	0.6	15.3	2	9.3–11.0	10.1	**10.1**	1.2	11.8
3	*Ateles*	2.3–3.9	2.9	**3.1**	0.8	26.3	3	6.4–8.7	6.8	**7.3**	1.3	17.3
2	*Alouatta*	1.3–1.6	1.4	**1.4**	0.2	16.0	2	2.7	2.7	**2.7**	0.0	0.4

Table 4 (continued)

		Brain volume						Cerebellum volume				
Number	Genus	Range cm^3	Median cm^3	Mean cm^3	SD cm^3	CV %	Number	Range cm^3	Median cm^3	Mean cm^3	SD cm^3	CV %
2	*Cebus*	2.2–2.5	2.4	**2.4**	0.2	7.1	2	4.6–4.8	4.7	**4.7**	0.2	3.5
2	*Saimiri*	0.9–1	1.0	**1.0**	0.1	12.5	2	0.9–1.0	1.0	**1.0**	0.1	7.8
4	*Aotus*	0.8–.95	0.9	**0.9**	0.1	7.5	4	0.8–1.2	0.9	**1.0**	0.2	15.8
	Ranges, medians, means, standard deviations, and coefficients for grades based on Genus/Species means above											
8	Human	9.5–15.7	10.9	**11.7**	2.4	20.3	8	114.6–166.3	125.4	**129.0**	15.9	12.3
14	Great ape	3.4–7.8	5.5	**5.5**	1.1	19.4	14	18.8–52.9	44.0	**42.2**	9.8	23.2
5	Lesser ape	2–2.6	2.3	**2.3**	0.3	14.0	5	8.6–12.2	9.9	**10.3**	1.5	14.5
8	OW monkeys	2.2–4.4	3.2	**3.3**	0.6	19.0	8	3.3–11.0	6.4	**6.6**	2.6	40.2
13	NW monkeys	0.8–3.9	1.4	**1.7**	0.9	54.6	13	0.8–8.7	2.7	**3.3**	2.7	81.4
	Average coefficients of variation for sample volumes											
	%											
	Whole Brain		11.8									
	Cerebellum		11.3									
	Hemisphere		12.4									
	Vermis		20.3									

B. Yerkes ranges, medians, means, standard deviations (SD), and coefficients of variation (CV)

		Brain volume						Cerebellum volume				
Number	Genus	Range cm^3	Median cm^3	Mean cm^3	SD cm^3	CV %	Number	Range cm^3	Median cm^3	Mean cm^3	SD cm^3	CV %
6	*Homo*	1088–1426	1332	**1298.8**	127.5	9.8	6	111–154	144.5	**137.0**	17.3	12.6
2	*Gorilla*	331–464	397.3	**397.3**	94.2	23.7	2	56–81	68.7	**68.7**	17.7	25.9
4	*Pan paniscus*	281–332	315.8	**311.2**	21.9	7.0	4	40–51	42.9	**44.2**	5.7	12.9
7	*Pan troglodytes*	283–391	322.4	**334.5**	36.1	10.8	7	40–52	46.1	**47.1**	4.3	9.2
4	*Pongo*	343–478	403.5	**406.9**	57.5	14.1	4	43–59	48.6	**49.8**	7.5	15.0
4	*Hylobates*	76–100	78.2	**83.0**	11.3	13.6	4	9–13.6	12.0	**11.6**	2.3	19.8
5	*Macaca*	73–100	84.4	**84.9**	9.6	11.3	6	6–8.3	7.6	**7.2**	0.9	12.4
4	*Cercocebus*	96–104	97.9	**98.9**	3.2	3.2	4	7.4–10.6	8.3	**8.7**	1.4	15.9
2	*Papio*	127–159	143.3	**143.3**	22.5	15.7	2	12.4–14.8	13.6	**13.6**	1.7	12.5
4	*Cebus*	55–77	66.7	**66.5**	10.5	15.8	4	5.1–7.7	6.5	**6.5**	1.4	21.5
4	*Saimiri*	21–24	23.7	**23.1**	1.6	6.8	4	1.93–2.2	2.0	**2.0**	0.1	5.7
	Average range, median, mean, standard deviation, and coefficient for grades											
6	Human	1088–1426	1332	**1298.8**	127.5	9.8	6	111–154	144.5	**137.0**	17.3	12.6
17	Great ape	281–478	331.8	**362.5**	46.9	12.9	17	40–81	46.1	**52.4**	11.1	21.1
4	Lesser ape	76–100	78.2	**83.0**	11.3	13.6	4	9–13.6	12.0	**11.6**	2.3	19.8
11	OW monkey	73–159	97.8	**109.0**	30.5	13.6	12	6–14.8	7.9	**9.8**	3.3	34.0
8	NW monkey	21–77	39.8	**44.8**	30.7	68.5	8	1.93–7.7	3.6	**4.3**	3.1	73.3

Table 4 (continued)

		Brain volume						Cerebellum volume				
Number	Genus	Range cm^3	Median cm^3	Mean cm^3	SD cm^3	CV %	Number	Range cm^3	Median cm^3	Mean cm^3	SD cm^3	CV %
		Vermis volume						Hemisphere volume				
Number	Genus	Range cm^3	Median cm^3	Mean cm^3	SD cm^3	CV %	Number	Range cm^3	Median cm^3	Mean cm^3	SD cm^3	CV %
6	*Homo*	9.3–15.0	11.6	**11.8**	2.3	19.8	6	102.1–139.0	131.6	**125.2**	15.3	12.2
2	*Gorilla*	4.9–9.1	7.0	**7.0**	3.0	42.5	2	51.2–72.1	61.6	**61.6**	14.8	24.0
4	*Pan paniscus*	4.8–9.6	5.5	**5.3**	0.7	13.7	4	34.7–41.8	37.4	**37.8**	3.0	8.0
7	*Pan troglodytes*	4.0–6.1	4.9	**5.0**	0.6	12.9	7	35.6–46.9	41.4	**42.1**	4.2	9.9
4	*Pongo*	4.7–6.1	5.0	**5.2**	0.6	12.2	4	38.0–53.2	43.6	**44.6**	6.9	15.5
4	*Hylobates*	1.7–2.4	1.9	**1.9**	0.3	15.1	4	7.2–11.7	9.9	**9.7**	2.1	21.6
6	*Macaca*	1.8–3.0	2.3	**2.3**	0.5	21.6	6	4.1–5.4	5.2	**4.9**	0.5	10.2
4	*Cercocebus*	1.9–2.9	2.2	**2.3**	0.4	18.5	4	5.5–7.7	6.1	**6.4**	1.0	15.2
2	*Papio*	3.3–3.9	3.6	**3.6**	0.4	10.4	2	9.1–10.9	10.0	**10.0**	1.3	13.2
4	*Cebus*	1.4–2.2	1.8	**1.8**	0.4	19.4	4	3.7–5.7	4.6	**4.6**	1.1	23.2
4	*Saimiri*	0.8–1.0	0.9	**0.9**	0.1	11.1	4	1.1–1.2	1.2	**1.2**	0.1	4.9
	Average range, median, mean, standard deviation, and coefficient for grades											
6	Human	9.3–1.5	11.6	**11.8**	2.3	19.8	6	102.1–139.0	131.6	**125.2**	15.3	12.2
17	Great ape	4.0–9.6	4.9	**5.3**	1.2	21.6	17	35.6–72.1	41.4	**44.0**	9.3	21.2
4	Lesser ape	1.7–2.4	1.9	**1.9**	0.6	32.9	4	7.2–11.7	9.9	**9.7**	2.1	21.6
12	OW Monkey	1.8–3.9	2.3	**2.5**	0.1	2.5	12	4.1–10.9	5.5	**6.2**	2.0	32.4
8	NW Monkey	0.8–2.2	1.2	**1.3**	0.2	13.3	8	1.1–5.7	2.5	**2.9**	2.0	68.5
	Average coefficient of variation for sample volumes											
			%									
	Whole brain		12.0									
	Cerebellum		14.9									
	Hemisphere		14.4									
	Vermis		17.9									

Table 5
Hirnforschung and Yerkes combined means

Genus	Whole brain volume			Cerebellum volume			Vermis volume			Hemisphere volume		
	CV Hirn and Yerkes %	Trimmed combined means cm^3	SEM combined means cm^3	CV Hirn and Yerkes %	Trimmed combined means cm^3	Sem combined means cm^3	CV Hirn and Yerkes %	Trimmed combined means cm^3	SEM combined means cm^3	CV Hirn and Yerkes %	Trimmed combined means cm^3	SEM combined means cm^3
Homo	3.3	**1265.1**	40.1	1.9	**139.2**	4.5	0.6	**11.8**	0.60	2.1	**127.4**	4.5
Gorilla	1.2	**401.3**	24.1	17.9	**59.4**	5.8	12.8	**6.3**	0.84	18.5	**53.1**	5.1
Pan paniscus	7.4	**322.7**	13.1	13.5	**47.3**	2.7	17.6	**6.5**	0.76	14.9	**40.8**	2.2
Pan troglodytes	8.1	**361.7**	13.6	1.0	**48.2**	1.4	3.1	**5.2**	0.34	3.5	**42.6**	1.3
Pongo	14.0	**391.6**	19.9	16.7	**47.7**	2.9	14.9	**5.0**	0.33	18.2	**43.8**	2.6
Hylobates	11.7	**91.4**	4.9	6.0	**12.2**	0.6	12.5	**2.2**	0.11	4.5	**10.0**	0.6
Erythrocebus	–	**89.0**	–	–	**7.9**	–	–	**2.7**	–	–	**5.2**	–
Macaca	14.8	**90.6**	5.2	20.0	**7.8**	0.5	22.4	**2.5**	0.21	18.5	**5.3**	0.3
Cercopithecus	–	**64.0**	7.1	–	**6.2**	0.6	–	**2.7**	0.50	–	**3.5**	0.2
Cercocebus	8.8	**101.5**	2.9	20.6	**9.3**	0.8	37.7	**2.6**	0.83	13.5	**6.6**	0.5
Papio	1.5	**144.8**	6.8	2.3	**13.8**	0.7	6.4	**3.8**	0.23	0.8	**10.1**	0.5
Ateles	–	**90.7**	10.3	–	**10.4**	0.9	–	**3.1**	0.50	–	**7.3**	0.7
Alouatta	–	**37.4**	0.9	–	**4.1**	0.2	–	**1.4**	0.20	–	**2.7**	0.01
Cebus	12.2	**69.0**	4.8	8.2	**6.6**	0.6	21.9	**1.9**	0.19	2.8	**4.7**	0.4
Saimiri	10.7	**22.0**	0.9	3.1	**2.0**	0.0	5.4	**0.9**	0.04	9.5	**1.1**	0.05
Aotus	–	**17.4**	1.3	–	**1.7**	0.6	–	**0.8**	0.03	–	**0.9**	0.07
Grades												
Humans	3.3	**1265.1**	40.1	1.9	**139.2**	4.5	0.6	**11.8**	0.60	2.1	**127.4**	4.5
Great apes	0.4	**369.3**	17.7	6.1	**50.6**	2.9	6.3	**5.7**	0.38	3.0	**43.8**	2.7
Lesser apes	11.7	**91.4**	4.9	6.0	**12.2**	0.6	12.5	**2.2**	0.11	4.5	**10.0**	0.6
OW monkeys	3.9	**98.0**	13.2	8.6	**9.0**	1.3	15.7	**2.9**	0.23	3.7	**6.1**	1.1
NW monkeys	6.1	**47.3**	14.1	12.6	**5.0**	1.6	11.7	**1.6**	0.41	10.0	**3.3**	1.2

NOTE: Some Hirnforschung species are unknown, in the cases of *Cercopithecus, Ateles, and Alouatta.*

Trimmed combined means—Two specimens from the Hirnforschung sample, Pongo 0475 and Chimpanzee 1548, were dropped from the absolute means because they were 3 and 2 standard deviations respectively from the species means.

They were not dropped from the regression tables as they did not present themselves as outliers when scaled allometrically.

CV Hirnforschung and Yerkes—Means from common species are compared.

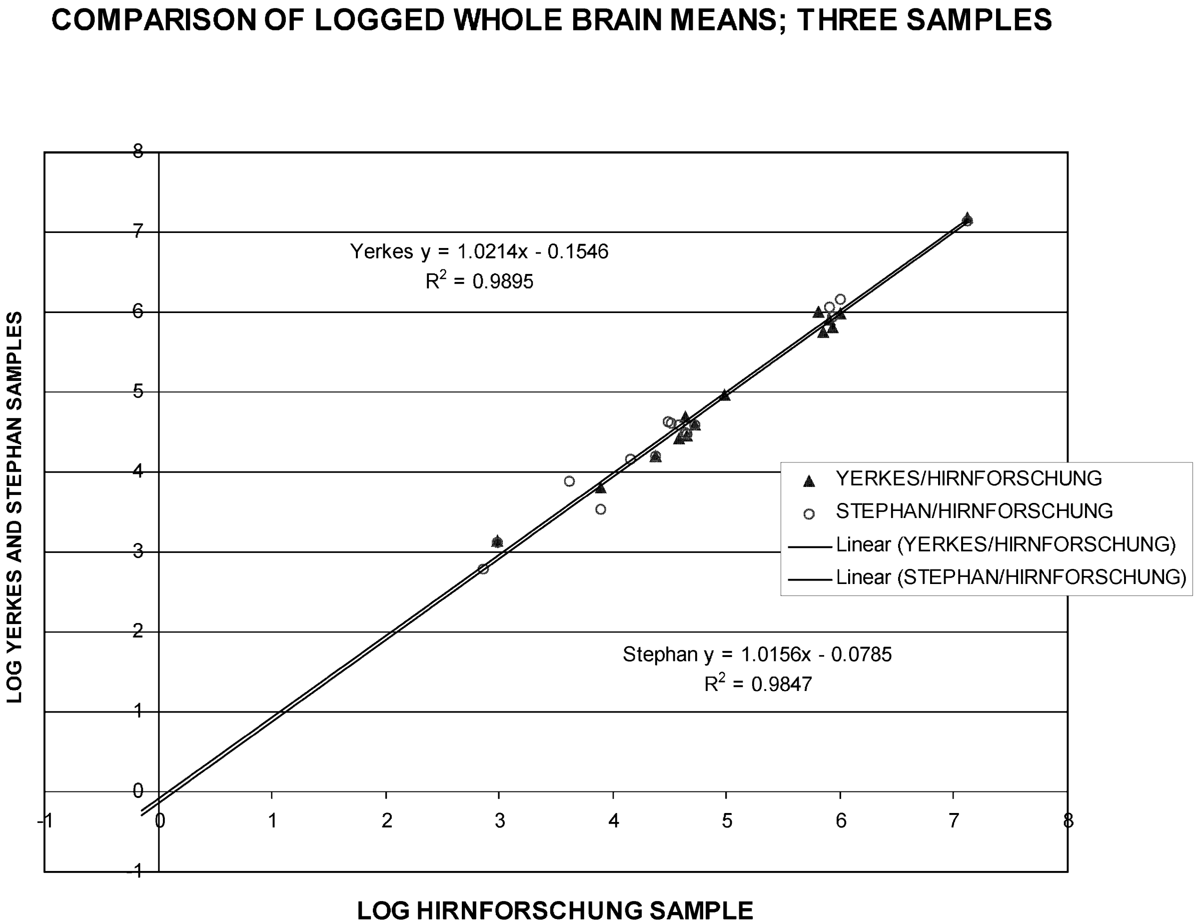

Fig. 5. The Yerkes and Stephan species means regressed against the Hirnforschung species means for logged whole brain volume. When the unlogged Hirnforschung and Yerkes species volumes are compared in a paired t-test, $p = 0.95$.

apes. Even though gibbons do not group with great apes in absolute values, they do group with other apes in proportions. Old World and New World monkeys are not significantly different from one another. Humans do not group with the rest of the hominoids, and are significantly different from Old World monkeys but not from New World monkeys ($p = 0.151$). These ratios illustrate that, contrary to Clark et al., (2001), the percentage of cerebellum to whole brain can vary by more than 2% above or below the figure of 13% within the primate order (see the discussions by Barton, 2002; Sultan, 2002; Wang, 2002 in *Nature*, 2002).

Taxonomic distinctions become even more apparent when the hemispheres are isolated from the rest of the cerebellum and compared to total brain volume (Table 6). The hemispheres occupy 10.1% of the brain in humans, 11.9% in the great apes, 11% in gibbons, and only 6.3% in OW monkeys and 6.1% in NW monkeys. Humans, great apes and gibbons are significantly different from monkeys when Tukey's test is performed. Old World and New World monkeys are not significantly different from each other ($p = 0.997$). Gibbons are not significantly different from great apes ($p = 0.378$) or humans ($p = 0.587$) in hemisphere to whole brain ratios.

In both of these cerebellar and hemisphere ratios, gibbons group with the other apes. Within the cerebellum, however, the vermis of the gibbon

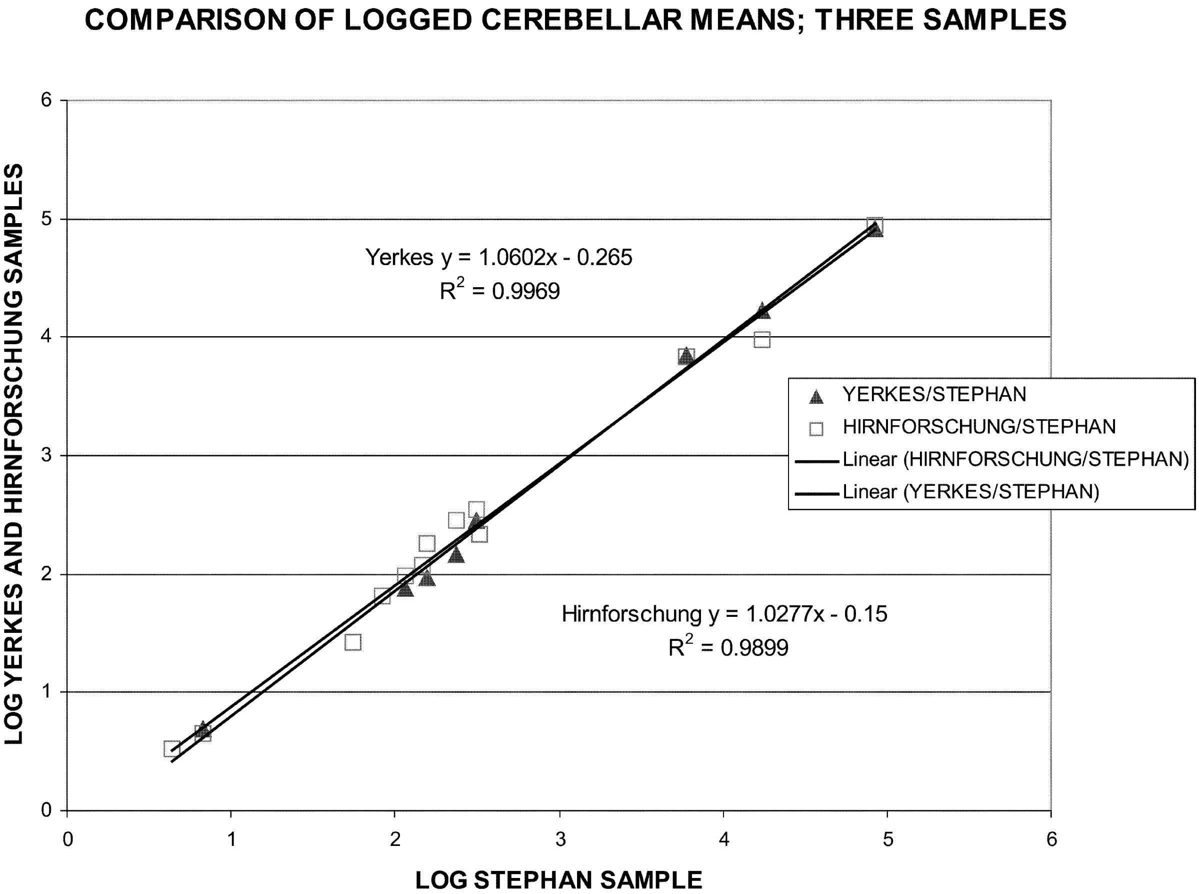

Fig. 6. Yerkes and Hirnforschung means regressed against Stephan means for logged cerebellar volumes. When the unlogged species means are compared between the Yerkes and Hirnforschung sets, p = 0.79.

Table 6
Ratios, SD, and SEM

Cerebellum to whole brain				Hemisphere to whole brain			Vermis to cerebellum		
Grade	Mean	SD	SEM	Mean	SD	SEM	Mean	SD	SEM
Human	11.0	0.83	0.22	10.0	0.74	0.2	8.4	1.09	0.29
Great ape	13.4	1.95	0.34	11.9	1.75	0.32	11.5	2.35	0.42
Lesser ape	13.4	1.92	0.64	11.0	1.8	0.6	18.0	2.48	0.83
OW monkey	9.2	0.82	0.19	6.3	0.69	0.16	31.4	5.47	1.25
NW monkey	9.8	1.21	0.26	6.1	1.44	0.32	38.2	9.04	1.97

occupies proportionately more of total cerebellar volume. Regression analysis shows the vermis to be expanding at a lower exponent than the hemispheres (Table 7), which could account for the higher gibbon vermis ratio in a cerebellum that is much smaller than the rest of the hominoids, i.e., as volume increases, those structures with higher exponents such as the hemispheres come to occupy

Table 7
Hemisphere to vermis cerebellum, hemispheres, and vermis to whole brain minus cerebellum (full model with interaction)

Regression	r^2 val.	SE	Derived formulae	Grade Sig.	Parallel
Hemisphere to vermis	0.968	0.268	y′ (monkeys) = 0.367 + 1.458x y′ (hominoids) = 1.465 + 1.365x	yes	yes
Cerebellum to whole brain minus cerebellum	0.988	0.147	y′ (monkeys) = −2.289 + 1.008x y′ (hominoids) = −1.3831 + 0.906x	yes	no
Hemisphere to whole brain minus cerebellum	0.988	0.165	y′ (monkeys) = −3.341 + 1.160x y′ (hominoids) = −1.731 + 0.945x	yes	no
Vermis to whole brain minus CB	0.938	0.194	y′ (monkeys) = −2.242 + 0.722x y′ (hominoids) = −1.990 + 0.632x	no	yes

more of the total cerebellar volume. Hence, the vermis occupies 31.4% of the cerebellum in Old World monkeys, 38.2% in New World monkeys, 18% in lesser apes, 11.5% in great apes and only 8.4% of total cerebellar volume in humans (Table 6). All grades are significantly different from each other except for great apes and humans (p = 0.375). Even with the larger vermis in gibbons over great apes, however, hominoid cerebella are dominated by the hemispheres, and appear quite distinct from those of monkeys.

Regression analysis

When the data are subject to regression analysis, these differences in proportion reflect not only a higher exponent of increase of hemisphere over vermis, but an actual differential expansion of the cerebellar hemispheres beyond expected allometry. Within the cerebellum, the hemispheres of hominoids are 2.7 times the size that would be expected in a monkey cerebellum of a given vermis volume. Both gibbons and humans scale closely with the rest of the hominoids in a clear grade shift of hominoid over monkey (Fig. 7).

To see whether this increase in cerebellar hemisphere to vermis ratio can explain the differential increase in the size of the cerebellum to the rest of the brain in hominoids, as reported previously (Rilling and Insel, 1998) and verified in the present study, the cerebellar hemispheres were regressed against the whole brain minus the cerebellum (Fig. 8). This too showed a grade shift between hominoids and monkeys, one that was more pronounced than that of the cerebellum to the rest of the brain (Table 7). When the vermis is regressed against the whole brain minus the cerebellum, there is a negative shift in the hominoid vermis volume. The shift is not significant, however, and data are best explained by a single regression line. The low slope of the vermis indicates that it is not increasing vigorously relative to the rest of the brain (Fig. 9). Thus, the differential expansion of the cerebellum in hominoids can be primarily attributed to a dramatic enlargement in the size of a specific part of the cerebellum, the hemispheres. These regressions are summarized in Table 7.

Both the cerebellum and the hemispheres show a positive grade shift in hominoids when regressed against the rest of the brain. The r^2 values are very high, and the standard error low, showing a good fit. However, these two regressions differ from the fundamental hemisphere to vermis relation because the slopes of hominoid and monkey are not parallel. Humans fit with the other hominoids in the internal cerebellar proportions, but are below expected values when regressed against the rest of the brain as a whole (see Semendeferi and Damasio, 2000). They are at one extreme of a very large range of distribution, and exercise leverage, pulling the slope of the hominoid lower. In both regressions, humans are not significantly different from apes, but their separation from apes restores the symmetry of parallel regression lines. Thus, the hominoid slope itself is not decreasing and showing a lower rate of expansion of the cerebellum and its hemispheres, but humans are below expected values. Rilling and Insel, (1998, 1999) suggest that

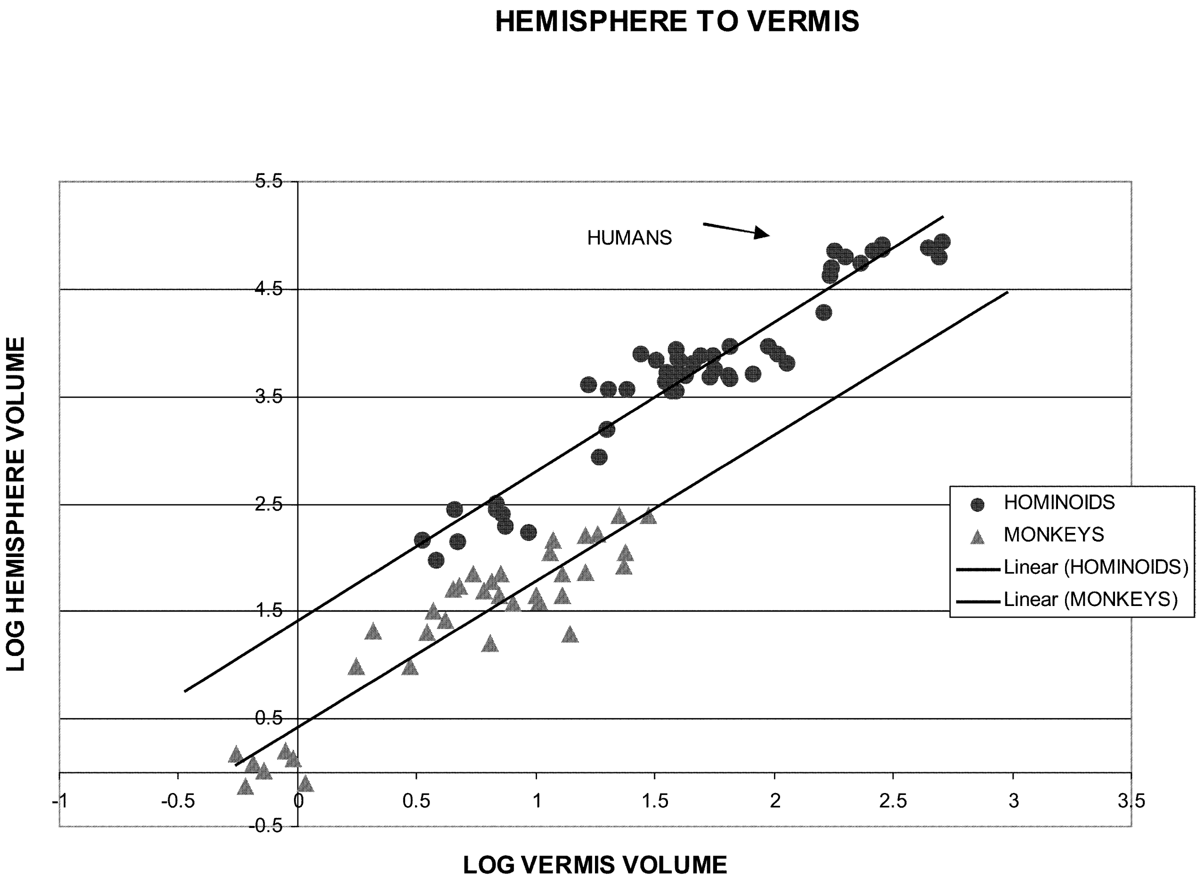

Fig. 7. Double logarithmic plot showing hemisphere to vermis proportions. Double regression lines show the grade shift of hominoids over monkeys. Regression formula from Table 7.

this reflects a change in the denominator because of a differential expansion of the neocortex in humans. In as yet unpublished data from the Institute für Hirnforschung, the first author has confirmed this hypothesis. When humans are given expected neocortical size determined from the anthropoid regression line, and whole brain size is adjusted accordingly, humans have cerebellar and cerebellar hemisphere volumes that are exactly as predicted by the ape regression line.

When humans are removed from the regression, and symmetry restored, it is possible to calculate the distance between the monkey and ape slopes by substituting the median x-value for the two equations, then restoring the logs to their natural numbers and calculating their ratio (Table 8). This method is more precise than calculating the difference in the intercepts of the two regression lines because the two slopes are not exactly parallel. By this method, apes show a positive grade shift over monkeys in cerebellar hemisphere expansion of 73%.

When the Stephan primate data set is combined with the Yerkes and Hirnforschung data set (N = 140), and whole brain volume regressed against body weight[3] , the ape has a brain volume

[3] When body weights were not available for individual specimens, or if weights were extremely low because of immature specimens, the Stephan body weights were used for the Yerkes and Hirnforschung samples.

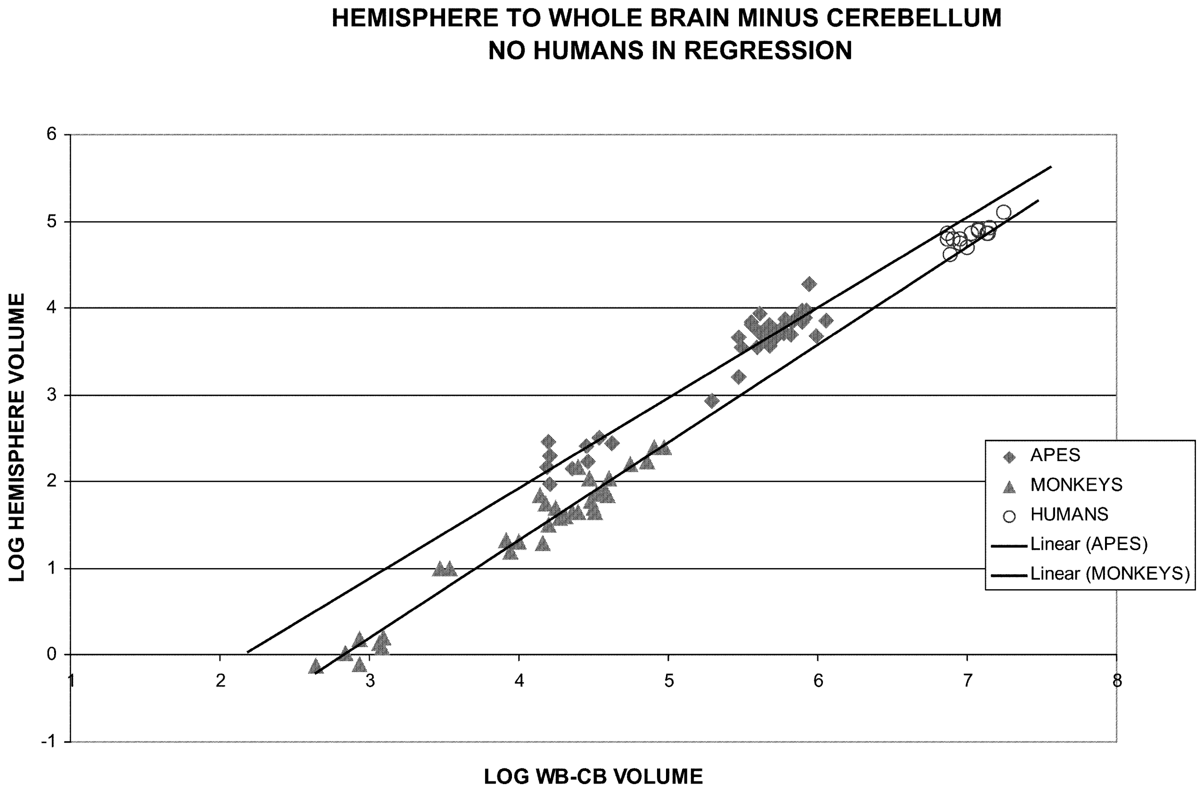

Fig. 8. Double logarithmic plot of hemispheres regressed against the whole brain minus cerebellum. Regression from Table 8. Humans are removed from the regression line (see text).

that is 40% larger than a monkey of the same body weight (Table 9). Humans must be excluded from this regression because they are so highly encephalised and distort the regression line for hominoids. The cerebellar hemispheres regressed against the body (Yerkes and Hirnforschung data sets only), show a differential expansion of 2.4 times relative to monkey hemispheres. Even with the high degree of error in brain to body calculations due to uncontrolled variation in body weight, the symmetry of the grade shift for the cerebellar hemispheres remains, showing the power of allometric scaling (Figs. 10 and 11).

Body weights for humans were rarely available in the two samples, so an average weight of 65 kg was substituted for x in the ape and monkey regression lines above in order to determine the magnitude of the human residuals. The expected brain and cerebellar hemisphere volumes were then divided into the actual mean human volume of 1265 cm^3 to determine the nature of human brain expansion.

Based on the combined Stephan, Yerkes and Hirnforschung data bases, humans have brains that are 3.5 times that expected in an ape of 65 kg, and 4.8 times that of an equivalently sized monkey. The cerebellar hemispheres in humans are 2.9 times the expected ape value, and 5.8 times the expected size in a monkey of 65 kg (Table 10). Thus, although humans are highly encephalised, this should not be taken with respect to the neocortex only. For our body weight, we have disproportionately large cerebellar hemispheres compared to apes, and even more so, compared to monkeys. In this larger perspective, the dramatic evolution of the cerebellum, especially its hemispheres, is an important aspect of the evolution of the human brain.

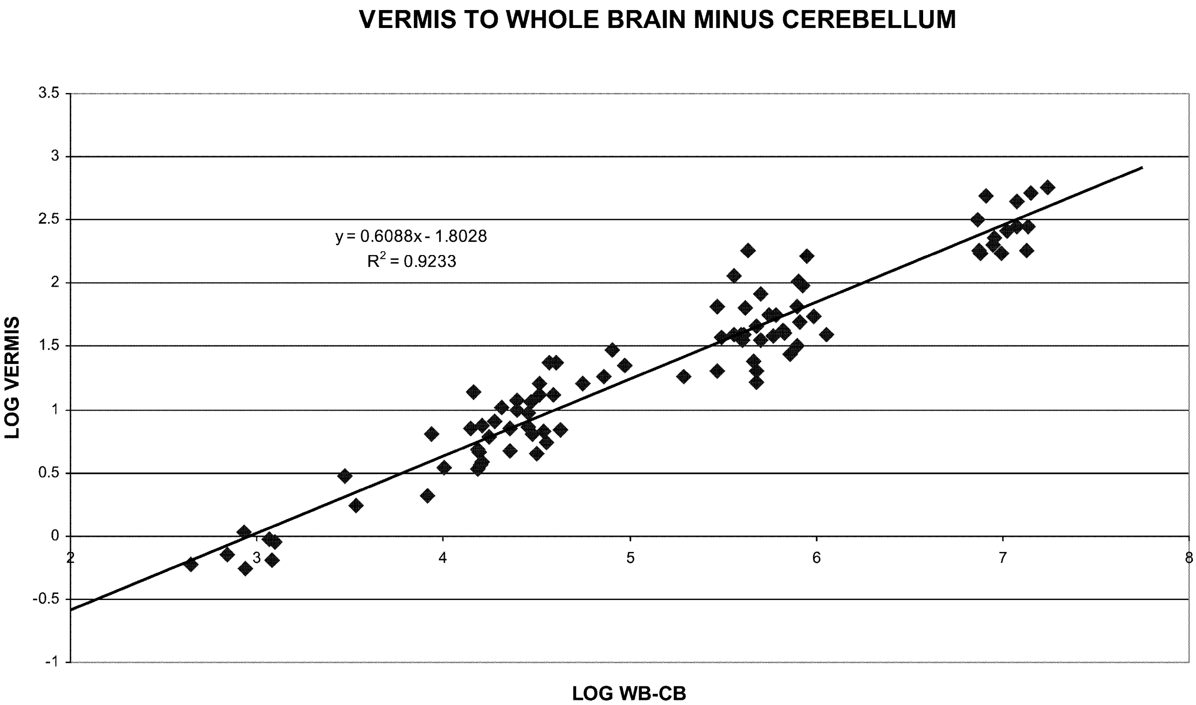

Fig. 9. The linear relation between the vermis and the whole brain minus cerebellum is best explained by a single regression line. There is no significant grade shift between apes and hominoids.

Table 8
Regression formulae for cerebellar hemispheres and cerebellum to whole brain minus cerebellum volume

Regression	r^2 value	SE	Derived formulae	Grades different	Parallel slopes	Predicted y (antilogs in cm^3)
Cerebellar hemispheres	0.984	0.167	y' (monkey) = $-3.341 + 1.160x$	yes	yes	10.9
			y' (ape) = $-2.334 + 1.061x$	–	–	18.9
Cerebellum	0.983	0.150	y' (monkeys) = $-2.289 + 1.008x$	no	yes	14.7
			y' (ape) = $-1.843 + 0.995x$	–	–	21.6

Full model with interaction. No humans in regression equation.

Independent contrasts

When CAIC is applied to the data to examine the differential expansion of the lateral cerebellum in hominoids, and to test whether the y-values can be predicted with equal certainty from the x-values when the data are analyzed according to phylogenetic contrasts, the original analysis based on multiple regression is confirmed. Table 11 shows the statistics for the regression of the standardized contrasts of x and y values for hemisphere to vermis, and hemisphere and cerebellum to whole brain minus cerebellum. When the standardized contrasts (raw contrasts divided by square root of expected variances) for y are regressed against those for x and the regression forced through the origin (0), a high correlation is found between the x and y contrasts.

Table 9
Regression formulae for whole brain and cerebellar hemisphere volumes to body mass

Regression	r^2 value	SE	Derived formulae	Grades different	Parallel slopes	Predicted y (antilogs in cm^3)
Whole brain	0.956	0.296	y′ (prosimian) = 2.799 + 0.761x	yes: pro./monk.	no: pro./monk.	–
			y′ (monkey) = 3.314 + 0.542x	yes: monk./ape	yes: monk./ape	83.7
			y′ (ape) = 3.725 + 0.521x	–	–	120.9
Cerebellar hemisphere	0.95	0.297	y′ (monkey) = 0.354 + 0.650x	yes	yes	6.8
			y′ (ape) = 1.455 + 0.553x	–	–	16.1

Full model with interaction. No humans in regression equation.

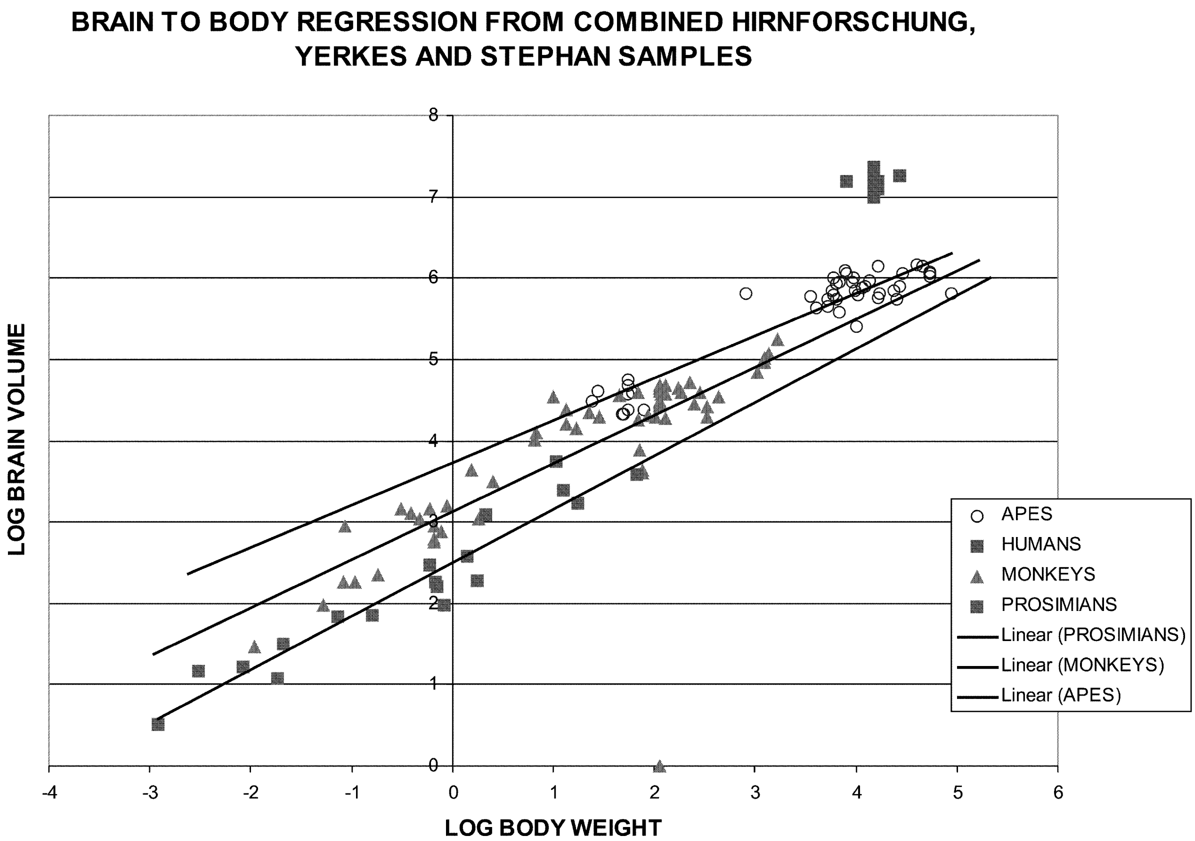

Fig. 10. Double logarithmic plot of brain to body regression for three data bases. Humans are removed from the regression (Table 9).

The slope for the hemisphere to vermis relation is steeper than when determined through multiple regression and two grades. This discrepancy is largely a phenomenon of leverage at both ends of the regression line. Longer branch lengths will understandably have more range of variability, whereas slight differences in x and y relations in recently evolved species will produce large residuals when the standardized contrasts for x and y are regressed on one another. For example, the

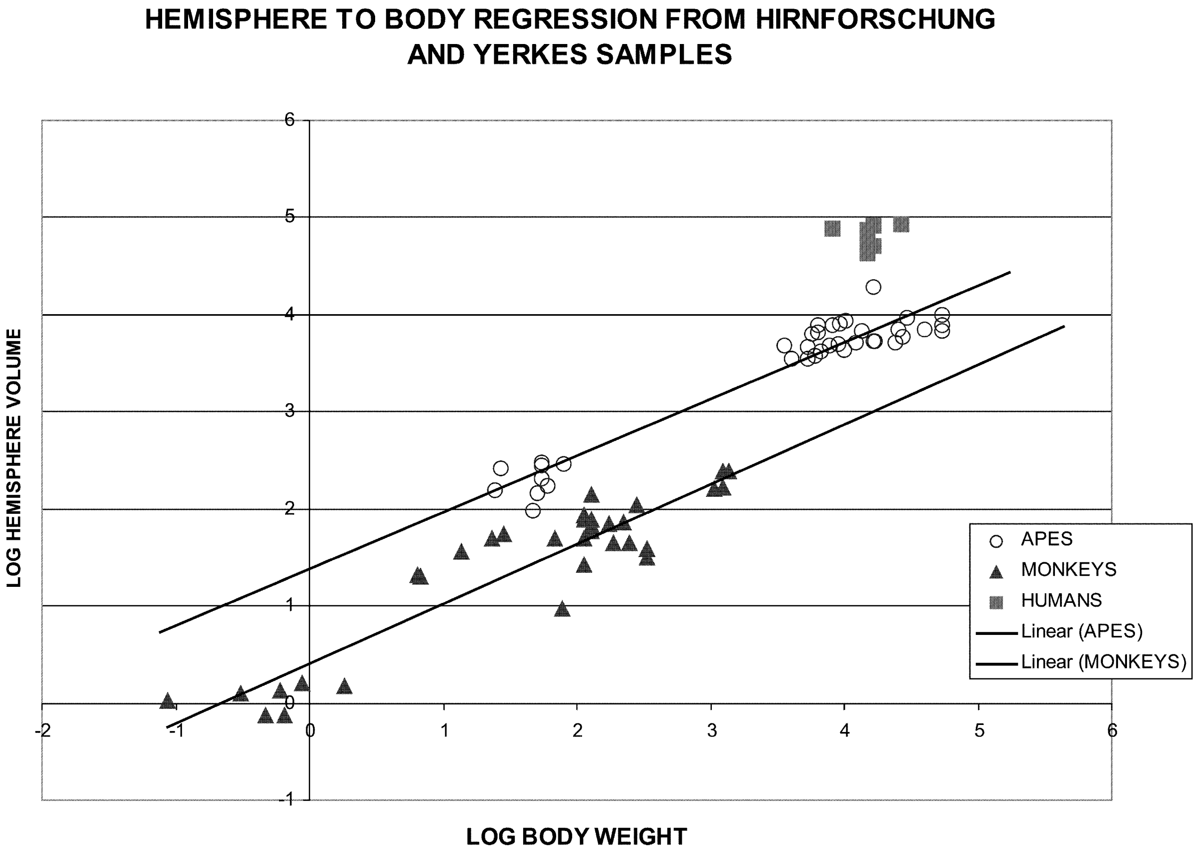

Fig. 11. Double logarithmic plot of cerebellar hemispheres to body weight. No humans in regression (equation from Table 9).

contrast within the genus *Pan* produces a negative residual in the hemisphere to vermis relation because the sample of *Pan paniscus* has a smaller hemisphere to a vermis proportion than *Pan troglodytes*. Since the two species diverged relatively recently, the residual is more pronounced than if those same differences in x and y values were found in an older lineage. This can be problematic when there is such a range of biological variability in the volumes of brain structures within a species or a genus, as the differences in taxa are magnified by their shorter branch lengths, and these differences may not be at all significant. Purvis and Rambaut, (1995) acknowledge that the method is sensitive to "noise" at the branch ends because of measurement "error". Furthermore, the earlier splits are closer to the origin when graphed with linear regression, and the oldest divergence farthest from the origin. Thus, if the chimpanzee/bonobo contrast is removed from the regression line, the slope flattens from 1.7 to 1.6. If the Catarrhine/Platyrrhine contrast is removed because of the oldest divergence exercising leverage on the slope, the regression line flattens even further to 1.5, compared to the slope of approximately 1.5 for monkeys and 1.4 for hominoids obtained from multiple regression of the hemisphere to vermis (Table 7). Thus, the method of independent contrasts is subject to the vagaries of leverage when the y-contrasts are regressed against the x-contrasts, and the determination of a definitive, precise slope remains elusive.

In order to verify the grade shift between hominoids and monkeys, specifically between hominoids and Old World monkeys, the positive residual for the hominoid/OW monkey contrast

Table 10
Human residual values over the ape and monkey regression lines

Ape regression line	Monkey regression line
Whole brain to body residual	
y (apes) = 3.725 + (0.521*log65kg)	y (monkeys) = 3.314 + (0.542*log65kg)
unlogged y = 365.0 cm^3	unlogged y = 264.2 cm^3
Division: (1265/365=) **3.5**	Division: (1265/264.2=) **4.8**
Cerebellum to body residual	
y (apes) = 1.702 + (0.523*log65 kg)	y (monkeys) = 0.934 + (0.550*log65 kg)
unlogged y = 48.7 cm^3	unlogged y = 25.3 cm^3
Division: (139.2/48.7=) **2.9**	Division: (139.2/25.3=) **5.5**
Hemisphere to body residual	
y (apes) = 1.455 + (0.553*log65 kg)	y (monkeys) = 0.354 + (0.650*log65 kg)
unlogged y = 43.1 cm^3	unlogged y = 21.5 cm^3
Division: (124.3/43.1=) **2.9**	Division: (124.3/21.5=) **5.8**

Table 11
Regressions of independent contrasts

Contrast	r^2 value	Slope	F-ratio	p-value	Hominoid grade outlier
Hemisphere to Vermis	0.91	1.7002	128.23	<0.0001	yes
Cerebellum to WB minus CB	0.96	0.9797	370.93	<0.0001	yes
Hemisphere to WB minus CB	0.95	1.1003	259.71	<0.0001	yes

must be large enough to be considered an outlier, and the slopes of the two grades must be shown to be parallel through either ANCOVA or a t-test (Purvis and Rambaut, 1995). The independent contrasts were treated with the same multiple regression method described for the original treatment of the data, with the variable of "grade" dropped from the analysis (as the regression is forced through the origin eliminating a difference in the y-intercept), but the interaction of x_1 and x_2 retained to determine whether the slopes of the two grades are parallel. The slopes are not significantly different in the hemisphere to vermis regression ($p = 0.361$), and the cerebellum to rest of brain regression ($p = 0.144$). The slopes are not significantly different in the hemisphere to the rest of the brain only if humans are removed from the contrast, as would be expected ($p = 0.09$). All regressions through the origin display a consistent outlier with the contrast of hominoids to Old World monkeys, thus confirming the analysis from multiple regression of a hominoid grade shift in cerebellar proportions associated with the differential expansion of the cerebellar hemispheres.

When the method of CAIC is applied to the brain to body, cerebellum to body and hemisphere to body relations described in Table 9, the picture becomes clouded. There is not a clear grade shift between hominoids and Old World monkeys, especially with the gorilla contrast showing such a negative residual. Although the contrast between hominoids and OW monkeys is a positive residual, it is not an outlier, and so the criterion of a positive grade shift is not met. CAIC is an excellent method for discerning species-specific patterns by "weighting" species' residual values according to phylogenetic history, but the underlying pattern of a grade shift in hemisphere to body proportions is lost in the patterning of species-specific encephalisation.

Discussion

The function of the cerebellum is very complex, participating in motor and sensory modalities with a cytoarchitecture and somatotopy that is quite distinct from that of the neocortex. At the very least, the increase in relative cerebellar proportions would contribute to the information-processing capacity of the brain, and enable a more complex level of response to the environment, something that most would consider a hallmark of hominoid behaviour. However, the close relationship between cerebellar structure and efferent information enables a further level of interpretation of the expansion of the cerebellum, because this expansion is primarily in the lateral cerebellum, and hence is largely neocerebellar. There are particular neuropsychological functions associated with the neocerebellum (Leiner et al., 1986, 1989, 1991, 1993, Schmahmann, 1991, 1996), and these have implications for hominoid behaviour and cognition.

All of the regressions performed, of which only a few are presented here, show a clear grade shift in lateral cerebellar proportions between hominoids and monkeys. The uniformity and symmetry of the shift would argue for this event to have taken place in the common ancestor to the hominoids; otherwise, species-specific change would tend to pull the grade lines away from parallel. If larger cerebellar hemispheres were selected, and such an increase could not be fortuitous given the high degree of allometric uniformity shown by the data, then what advantages did an augmented neocerebellar structure offer to the early hominoids?

The cerebellum has traditionally been associated with co-ordination of movement (Holmes, 1939), but it appears now that the neocerebellum is also associated with cognitive aspects of movement. The medial and anterior cerebellum are the loci for the *execution* of motor patterns, and the lateral cerebellum is the locus for the *planning* of these movements (Thach, 1997). If the cerebellum were expanding only to enable more refined locomotion, then the vermis, associated with balance, equilibrium and execution of movement, would be expected to increase in volume at a much greater rate than the data show. Instead, it may be new movement patterns that are augmented with neocerebellar function.

Gibbons move with amazing speed and grace in a three-dimensional space that affords more scope of movement and direction and patterning than quadrupedal branch runners. Povinelli and Cant, (1995) have observed that macaque locomotion is stereotyped compared to orangutan clambering, which requires the planning and execution of unusually flexible locomotor patterns. Clambering is shared amongst the great apes, and Povinelli and Cant suggest that it requires a consciousness on the part of the animal that enables it to “see” itself in a three-dimensional space. This development would have come with the possible increase in body size in the Miocene apes, when their bodies became too ponderous for agile, quick locomotion, and movement became a more conscious decision-making process.

At this point, there is no clear suspensory ancestor for the hominoids that emerges from the early Miocene, and the obvious anatomical adaptations to suspensory behaviour evident in all hominoids may be homoplasies in lesser and great apes (Larson, 1998). Only some of the early apes showed suspensory behaviour, but all showed evidence of more versatile locomotor abilities closer to those of spider monkeys and chimpanzees (Fleagle, 1999). The proconsulids of the early Miocene were frugivorous, however (Fleagle, 1999), and it is plausible that the increased motor versatility afforded by the lateral cerebellum could be used in more efficient feeding strategies of reaching for food both over and under branches.

Frugivory requires the animal to remember patchily distributed fruiting trees over a large range (Milton, 1981, Clutton-Brock and Harvey, 1979). The animal must make a mental map of resources, and have well developed visuo-spatial abilities. The lateral cerebellum has been implicated in visuo-spatial tasks having a strong cognitive component (Kim et al., 1994); the large number of fibres connecting the cerebellum with the parietal-occipital areas of the brain (Stein et al., 1987) would be an important neurological substrate to these visuo-spatial skills.

Extractive foraging requires the individual to remove the nutrients from matrices in which they are embedded or encased (Gibson, 1986, Parker and Gibson, 1977, 1979). Extractive foraging as practised by orangutans, chimpanzees and humans often requires a complex series of manipulations in which some steps are embedded in others. This has been termed hierarchical mental construction (Gibson, 1990, Greenfield, 1991, Gibson and Jessee, 1999). Procedural learning, or learning how to *do* a task, is necessary to successful extractive foraging, tool use and other complex behaviour routines (Gibson, 1999, Parker and Milbrath, 1993). The neocortex appears to play a major role in the mediation of hierarchical mental construction, but may do so in co-operation with the neocerebellum. The lateral cerebellum is implicated in procedural learning in a number of different animal species, from the learning of a Morris water maze by rats (Doyon, 1997) to the learning of how to generate verbs from nouns by humans (Fiez and Raichle, 1997, Raichle et al., 1994). Thus, increased lateral cerebellar cortex could have aided hominoids to exploit foods that required a learning component, and could, for example, help chimpanzees to crack open nuts or extract termites from mounds.

In the case of the mountain gorilla, food must be processed beyond extraction to remove barriers such as stings, prickles, hard casings or tiny hooks in a series of logical steps (Byrne, 1995). This requires dexterity and sensitivity. The lateral cerebellum has traditionally been associated with fine movements of the distal extremities, and Matano and Hirasaki, (1997) interpret their data on the neocerebellum as an augmentation of this function in humans, but many would see the lateral cerebellum as more of a sensory organ (Bower, 1997). In a functional magnetic resonance experiment, Gao and colleagues (1996) determined that there was greater activity in the lateral cerebellum when the subject was required to distinguish between shapes and textures of two different objects than when the subject was merely required to pick up and move the object from one place to the next. They suggest that the cerebellum acts as a sensory-motor interface by co-ordinating sensory feedback with motor exploration. The fine probings of the gorilla fingers in the process of feeding could be supported by this neocerebellar activity.

Other neocerebellar functions relevant to hominoid behaviour are being explored in cognitive psychology, especially those implicated in the ability to switch attention rapidly between visual and auditory modalities (Allen et al., 1997, Allen and Courchesne, 2001). This too would accord with the tendency in hominoids to have longer periods of infant dependancy, with larger repertoires of learned behaviour. In human language learning, the infant must know where to focus its attention in mother-infant interaction in order to learn context linkages (Tomasello and Todd, 1983, Tomasello and Farrar, 1986); the lateral cerebellum would enable hominoids to learn more easily by focusing on relevant contextual clues.

The importance of neocerebellar functions through the expansion of its chief output nucleus, the dentate, has been discussed by other researchers, notably Matano (1985b), and Matano and Hirasaki (1997). Results of the measurement of the principal inferior olive and the dentate will be reported in a subsequent paper, but they do not accord with their view that differential dentate expansion was a hallmark of human neocerebellar expansion. The increase in neocerebellar volume through the lateral cerebellar cortex increase is clear, supporting the idea that the neocerebellum is important to cognitive functions, especially in hominoids. The pattern of nuclear expansion, however, is not so straightforward (MacLeod et al., 2000, 2001).

An outline of neocerebellar functions with particular relevance to the early hominoid niche can, at this juncture, only suggest possible reasons for the selection of larger cerebellar hemispheres. Gibbons and humans show the same pattern of hemisphere to vermis expansion as the great apes, and although the *size* of the gibbon brain is far below that of the rest of the hominoids, its cerebellar proportions are clearly hominoid. Further work in neuropsychological testing of functions connected with the cerebellum might be one way to shed light on the question of a distinctive hominoid brain anatomy.

Summary

An expanded data set for the whole brain, the cerebellum and its components, reveals a marked increase in the size of the cerebellar hemispheres for hominoids over monkeys. This has implications in hominoids for a common set of cognitive abilities that may be partly dependent on an increased participation of the neocerebellum. The superior abilities of the great apes to acquire artificial languages under human tutelage, and their more complex ordering of actions in food processing and tool use, could be explained in part by the increase in cerebellar processing abilities, especially in the lateral cerebellum. This is not to deny the importance of size in cognitive ability, and the obvious advantage accrued to the large-brained great apes. However, the differential expansion of the cerebellar hemispheres are indicative of a new balance in neurological functioning with the apes and humans, a qualitative change in primate brain dynamics that likely took place in the common ancestor to Hominoidea. Although humans are unique in our enormous encephalisation, we nonetheless share this differential expansion of the lateral cerebellum with the apes. Our brain organisation has a significant cerebellar component, and the evolution of the human brain can be better understood with a recognition of its affinities with the brains of our closest relatives.

Acknowledgements

Supported by the Leakey Foundation, Simon Fraser University, the Langara Research Committee, and a fellowship from the Institute of Medicine, Forschungzentrum, Juelich, Germany. Thanks to the Yerkes Regional Primate Research Center, Atlanta, Georgia, and the Institut für Hirnforschung, Duesseldorf, Germany for invaluable institutional support. Thanks also to Murray Besler of the Mathematics Department of Langara College, to Shannon Wood for her technical help, to Professor Vito Modigliani of Simon Fraser University for his insight, and to two anonymous reviewers.

References

Aiello, L.C., 1992. Allometry and the analysis of size and shape in human evolution. J. Hum. Evol. 22, 127–147.

Akshoomoff, N.A., Courchesne, E., 1992. A new role for the cerebellum in cognitive operations. Behav. Neurosci. 106(5), 731–738.

Allen, G., Buxton, R.B., Wong, E.C., Courchesne, E., 1997. Attentional activation of the cerebellum independent of motor involvement. Science 275, 1940–1943.

Allen, G., Courchesne, E., 2001. Attention function and dysfunction in autism. Frontiers in Bioscience 6, 105–119.

Barton, R., 2002. How did brains evolve? Nature 415, 134–135.

Bauchot, R., Stephan, H., 1966. Données nouvelles sur l'encéphalisation des insectivores et des prosimiens. Mammalia 30, 160–196.

Bauchot, R., Stephan, H., 1969. Encéphalisation et niveau évolutif chez les simiens. Mammalia 33, 225–275.

Bower, J.M., 1997. Is the cerebellum sensory for motors sake, or motor for sensorys sake: The view from the whiskers of a rat, in: de Zeeuw, C.E., Strata, P., Voogd, J. (Eds.), The cerebellum: from structure to control. Elsevier, Amsterdam, pp. 463–498.

Byrne, R., 1995. The thinking ape; evolutionary origins of intelligence. Oxford University Press, Oxford.

Clark, D.A., Mitra, P.P., Wang, S.S.-H., 2001. Scalable architecture in mammalian brains. Nature 411, 189–193.

Clutton-Brock, T.H., Harvey, P.H., 1979. Home range size, population density and phylogeny in primates, in: Bernstein, I.S., Smith, E.O. (Eds.), Primate ecology and human origins. Garland Press, New York, pp. 201–214.

Courchesne, E., Townsend, J., Akshoomoff, N.A., Saitoh, O., Yeung-Courchesne, R.J., Lincoln, A.J., James, H.E., Haas, R.H., Schreibman, L., Lau, L., 1994. Impairment in shifting attention in autistic and cerebellar patients. Behav. Neurosci. 108(5), 848–865.

Doyon, J., 1997. Skill learning. Int. R. Neurobiol. 41, 273–294.

Fiez, J.A., Raichle, M.E., 1997. Linguistic processing. Int. R. Neurobiol. 41, 233–254.

Fleagle, J.C., 1999. Primate adaptation and evolution, Second edition Academic Press, Toronto.

Gao, J.-H., Parsons, L.M., Bower, J.M., Xiong, J., Li, J., Fox, P.T., 1996. Cerebellum implicated in sensory acquisition and discrimination rather than motor control. Science. 272, 545–547.

Gibson, K.R., 1986. Cognition, brain size and the extraction of embedded food resources, in: Else, J.G., Lee, P.C. (Eds.), Primate ontogeny, cognition and social behaviour. Vol. 3. Cambridge University Press, Cambridge, pp. 93–104.

Gibson, K.R., 1990. New perspectives on instincts and intelligence: Brain size and the emergence of hierarchical mental constructional skills, in: Parker, S.T., Gibson, K.R. (Eds.), 'Language' and intelligence in monkeys and apes; Comparative developmental perspectives. Cambridge University Press, New York, pp. 97–128.

Gibson, K.R., 1999. Social transmission of facts and skills in the human species: neural mechanisms, in: Box, H.O., Gibson, K.R. (Eds.), Mammalian social learning: Comparative and ecological perspectives (Symposia of the Zoological Society of London 72). Cambridge University Press, Cambridge, pp. 351–366.

Gibson, K., Jessee, S., 1999. Language evolution and expansions of multiple neurological processing areas, in: King, B.J. (Ed.), The origins of language; What nonhuman primates can tell us. School of American Research Press, Santa Fe, pp. 189–227.

Greenfield, P., 1991. Language, tools and brain: The ontogeny and phylogeny of hierarchically organized sequential systems. Behav. Brain Sci. 14(4), 531–595.

Harvey, P.H., Mace, G.M., 1982. Comparison between taxa and adaptive trends: problems of methodology, in: King's College Sociobiology Group (Ed.), Current problems in sociobiology. Cambridge University Press, Cambridge, pp. 343–361.

Holmes, G., 1939. The cerebellum of man. Brain 62, 1–30.

Howard, C.V., Reed, M.G., 1998. Unbiased stereology; Three-dimensional measurement in microscopy. BIOS Scientific Publishers, Oxford.

Huxley, J.S., 1958. Evolutionary processes and taxonomy with special references to grades. Uppsala Univ. Arsskr., 21–39.

Jansen, J., Brodal, A., 1954. Aspects of cerebellar anatomy. Johan Grundt Tanum Forlag, Oslo.

Kim, S.-G., Ugurbil, K., Strick, P.L., 1994. Activation of a cerebellar output nucleus during cognitive processing. Science. 265, 949–951.

Kleinbaum, D.G., Kupper, L.L., 1978. Applied regression analysis and other multivariable methods. Duxbury Press, North Scituate, Mass.

Larson, S.G., 1998. Parallel evolution in the hominoid trunk and forelimb. Evol. Anthrop. 6(3), 87–99.

Leiner, H.C., Leiner, A.L., Dow, R.S., 1986. Does the cerebellum contribute to mental skills? Behav. Neurosci. 100(4), 443–454.

Leiner, H.C., Leiner, A.L., Dow, R.S., 1989. Reappraising the cerebellum: What does the hindbrain contribute to the forebrain. Behav. Neurosci. 103(5), 998–1008.

Leiner, H.C., Leiner, A.L., Dow, R.S., 1991. The human cerebro-cerebellar system: its computing, cognitive, and language skills. Behav. Brain Res. 44, 113–128.

Leiner, H.C., Leiner, A.L., Dow, R.S., 1993. Cognitive and language functions of the human cerebellum. TINS 16(11), 444–447.

MacLeod, C.E., 2000. *The cerebellum and its part in the evolution of the hominoid brain.* PhD Dissertation, Simon Fraser University.

MacLeod, C.E., Zilles, K., Schleicher, A., Gibson, K.R., 2001. The cerebellum: An asset to hominoid cognition, in: Galdikas, B.M.F., Briggs, N.E., Sheeran, L.K., Shapiro, G.L., Goodall, J. (Eds.), All apes great and small, African apes. Volume I. Kluwer Academic/Plenum Publishers, New York, pp. 35–53.

Martin, R.D., 1980. Adaptation and body size in primates. Z. Morphol. Anthrop. 71, 115–124.

Martin, R.D., Harvey, P.H., 1985. Brain size allometry: Ontogeny and phylogeny, in: Jungers, W.L. (Ed.), Size and scaling in primate biology. Plenum Press, New York, pp. 147–174.

Matano, S., 1986. A volumetric comparison of the vestibular nuclei in primates. Folia Primat. 47, 189–203.

Matano, S., 1992. A comparative neuroprimatological study on the inferior olivary nuclei (from the Stephan's Collection). J. Anthrop. Soc. Nippon 100(1), 69–82.

Matano, S., Stephan, H., Baron, G., 1985a. Volume comparisons in the cerebellar complex of primates I. Ventral Pons. Folia Primatol. 44, 171–181.

Matano, S., Stephan, H., Baron, G., 1985b. Volume comparisons in the cerebellar complex of primates II. Cerebellar nuclei. Folia Primatol. 44, 182–203.

Matano, S., Hirasaki, E., 1997. Volumetric comparisons in the cerebellar complex of anthropoids, with special reference to locomotor types. Am. J. Phys. Anthrop. 103, 173–183.

Merker, B., 1983. Silver staining of cell bodies by means of physical development. J. Neurosc. Meth., 235–241.

Milton, K., 1981. Distribution patterns of tropical plant foods as an evolutionary stimulus to primate mental development. Am. Anthrop. 83, 534–548.

Molinari, M., Petrosini, I., Grammaldo, L.G., 1997. Spatial event pocessing, in: Schmahmann, J.D. (Ed.), The cerebellum and cognition. Academic Press, San Diego, pp. 217–230. (Int. R. Neurobiol. 41.)

Nunn, C.L., Barton, R.A., 2001. Comparative methods for studying primate adaptation and allometry. Evol. Anthrop. 10, 81–98.

Parker, S.T., Gibson, K.R., 1977. Object manipulation, tool use and sensorimotor intelligence as feeding adaptations in Cebus monkeys and great apes. J. Hum. Evol. 6, 623–641.

Parker, S.T., Gibson, K.R., 1979. A developmental model for the evolution of language and intelligence in early hominids. Behav. Brain Sci. 2, 367–408.

Parker, S.T., Milbrath, C., 1993. Higher intelligence, propositional language, and culture as adaptations for planning, in: Gibson, K.R., Ingold, T. (Eds.), Tools, language and cognition in human evolution. Cambridge University Press, Cambridge, pp. 314–333.

Povinelli, D., Cant, J.G.H., 1995. Arboreal clambering and the evolution of self-conception. Quar. R. Biol. 70(4), 393–421.

Purvis, A., 1995. A composite estimate of primate phylogeny. Phil. Trans. R. Soc. Lond. B 348, 405–421.

Purvis, A., Rambaut, A., 1995. Comparative analysis by independent contrasts (CAIC): an Apple Macintosh application for analysing comparative data. Computer Appl. Biosciences 11, 247–251.

Raichle, M.E., Fiez, J.A., Videen, T.O., MacLeod, A.-M.K., Pardo, J.V., Fox, P.T., Petersen, S.E., 1994. Practice-related changes in human brain functional anatomy during nonmotor learning. Cereb. Cortex 4, 8–26.

Rilling, J.K., Insel, T.R., 1998. Evolution of the cerebellum in primates: Differences in relative volume among monkeys, apes and humans. Brain Behav. Evol. 52, 308–314.

Rilling, J.K., Insel, T.R., 1999. The primate neocortex in comparative perspective using magnetic resonance imaging. J. Hum. Evol. 37, 191–223.

Schmahmann, J.D., 1991. An emerging concept: The cerebellar contribution to higher function. Arch. Neurol. 48, 1178–1187.

Schmahmann, J.D., 1996. From movement to thought: Anatomic substrates of the cerebellar contribution to cognitive processing. Hum. Brain Map. 4, 174–198.

Semendeferi, K., Damasio, H., 2000. The brain and its main anatomical subdivisions in living hominoids using magnetic resonance imaging. J. Hum. Evol. 38, 317–332. doi: 10.1006/jhev.1999.0381.

Stein, J.F., Miall, R.C., Weir, D.J., 1987. The role of the cerebellum in the visual guidance of movement, in: Glickstein, M., Yeo, C., Stein, J. (Eds.), Cerebellum and neuronal plasticity. Plenum Press, NY, pp. 175–192.

Stephan, H., Andy, O.J., 1969. Quantitative comparative neuroanatomy of primates: An attempt at a phylogenetic interpretation. Ann. N.Y. Acad. Sci. 167, 370–387.

Stephan, H., Bauchot, R., Andy, O.J., 1970. Data on size of the brain and of various brain parts in insectivores and primates, in: Noback, C.R., Montagna, W. (Eds.), The Primate Brain; Advances in Primatology. Vol. I. Appleton-Century-Crofts, New York, pp. 289–298.

Stephan, H., Frahm, H., Baron, G., 1981. New and revised data on volumes of brain structures in insectivores and primates. Folia primat. 35, 1–29.

Stephan, H., Frahm, H., Baron, G., 1988. Comparative size of brains and brain components, in: Steklis, H.D., Erwin, J. (Eds.), Comparative Primate Biology, Neurosciences. Vol. 4. Alan R. Liss, Inc., New York, pp. 1–38. pp. 1–38.

Sultan, F., 2002. Analysis of mammalian brain architecture. Nature 415, 133–134.

Talairach, J., Tournoux, P., 1988. Co-planar stereotaxic atlas of the human brain. Stuttgart, Thieme Verlag.

Thach, W.T., 1996. On the specific role of the cerebellum in motor learning and cognition: Clues from PET activation and lesion studies in man. Behav. Brain Sci. 19, 411–431.

Thach, W.T., 1997. Contex-response linkage, in: Schmahmann, J.D. (Ed.), The cerebellum and cognition. Academic Press, San Diego, pp. 599–611. (Int. R. Neurobiol. 41.)

Tomasello, M., Todd, J., 1983. Joint attention and lexical acquisition style. First Lang. 4, 197–212.

Tomasello, M., Farrar, M.J., 1986. Joint attention and early language. Child Dev. 57, 1454–1463.

Wang, S., 2002. *Clarke* et al. *reply*. Nature 415, 135.

Zilles, K., Schleicher, A., Pehlemann, F.-W., 1982. How many sections must be measured in order to reconstruct the volume of a structure using serial sections? Microsc. Act. 86(4), 339–346.